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JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Definite Integration

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 250 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

The value of the integral $\int_0^2 \frac{\sqrt{x(x^2+x+1)}}{(\sqrt{x+1})\left(\sqrt{x^4+x^2+1}\right)}\mathrm{d}x$ is equal to:

(A) $\frac{1}{3}\log_e(3-2\sqrt{2})$ (B) $\frac{2}{3}\log_e(4+\sqrt{2})$

(C) $\frac{2}{3}\log_e(3+2\sqrt{2})$ (D) $\frac{1}{3}\log_e(1+6\sqrt{2})$

Open solution

Q2. JEE Main 2026 (Online) 8th April Evening Shift

Let $f(x)=\begin{cases}\frac{1}{3}, & x\le \pi/2\\ \frac{\mathrm{b}(1-\sin x)}{(\pi-2x)^2}, & x>\pi/2\end{cases}$. If f is continuous at $x=\pi/2$, then the value of $\int_0^{3\,\mathrm{b}-6}|x^2+2x-3|\mathrm{d}x$ is :

(A) 5 (B) 2 (C) 3 (D) 4

Open solution

Q3. JEE Main 2026 (Online) 6th April Evening Shift

Let $\mathrm{A}=\begin{bmatrix}1 & 3 & -1\\ 2 & 1 & \alpha\\ 0 & 1 & -1\end{bmatrix}$ be a singular matrix. Let $f(x)=\int_0^x(\mathrm{t}^2+2\mathrm{t}+3)\mathrm{dt}, x\in[1,\alpha]$. If M and m are respectively the maximum and the minimum values of f in $[1,\alpha]$, then $3(\mathrm{M}-\mathrm{m})$ is equal to :

(A) 64 (B) 68 (C) 72 (D) 76

Open solution

Q4. JEE Main 2026 (Online) 6th April Evening Shift

The value of the integral $\int_{-1}^{1}\left(\frac{x^3+|x|+1}{x^2+2|x|+1}\right)\mathrm{d}x$ is equal to:

(A) $3\log_e 2$ (B) $2\log_e 2$ (C) $5\log_e 3$ (D) $3\log_e 3$

Open solution

Q5. JEE Main 2026 (Online) 6th April Morning Shift

The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{32\cos^4 x}{1+e^{\sin x}}\right)dx$ is :

(A) $4\pi+2$ (B) $3\pi+8$ (C) $3\pi+4$ (D) $4\pi+3$

Open solution

Q6. JEE Main 2026 (Online) 5th April Evening Shift

Let $\left(2^{1-\mathrm{a}}+2^{1+\mathrm{a}}\right), f(\mathrm{a}), \left(3^{\mathrm{a}}+3^{-\mathrm{a}}\right)$ be in A.P. and α be the minimum value of $f(\mathrm{a})$. Then the value of the integral $\int_{\log_e(\alpha-1)}^{\log_e(\alpha)}\frac{dx}{(e^{2x}-e^{-2x})}$ is :

(A) $\frac{1}{2}\log_e\left(\frac{4}{3}\right)$ (B) $\frac{1}{4}\log_e\left(\frac{4}{3}\right)$

(C) $\frac{1}{2}\log_e\left(\frac{8}{5}\right)$ (D) $\frac{1}{4}\log_e\left(\frac{8}{5}\right)$

Open solution

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Q7. JEE Main 2026 (Online) 5th April Morning Shift

The value of the integral $\int_0^{\infty} \frac{\log_e(x)}{x^2+4}dx$ is:

(A) $\frac{\pi \log_e(2)}{2}$ (B) $\frac{\pi \log_e(2)}{4}$ (C) $1+\pi \log_e(2)$ (D) $2+\pi \log_e(2)$

Open solution

Q8. JEE Main 2026 (Online) 5th April Morning Shift

The value of the integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left(\frac{4-\text{cosec}^2 x}{\cos^4 x}\right)dx$ is :

(A) $\frac{11}{\sqrt{3}}$ (B) $\frac{16}{\sqrt{3}}$ (C) $\frac{32}{3\sqrt{3}}$ (D) $\frac{64}{3\sqrt{3}}$

Open solution

Q9. JEE Main 2026 (Online) 4th April Evening Shift

If $\alpha = 1$ and $\beta = 1+i\sqrt{2}$, where $i = \sqrt{-1}$ are two roots of the equation $x^3+ax^2+bx+c=0, a, b, \in \mathbb{R}$, then $\int_{-1}^{1}(x^3+ax^2+bx+c)dx$ is equal to:

(A) -2 (B) -4 (C) -8 (D) -10

Open solution

Q10. JEE Main 2026 (Online) 4th April Evening Shift

The integral $\int_0^1 \cot^{-1}\left(1+x+x^2\right)dx$ is equal to :

(A) $2\tan^{-1}2+\frac{1}{2}\log_e\left(\frac{5}{4}\right)+\frac{\pi}{2}$ (B) $2\tan^{-1}2+\frac{1}{2}\log_e\left(\frac{5}{4}\right)-\frac{\pi}{2}$

(C) $2\tan^{-1}2-\frac{1}{2}\log_e\left(\frac{5}{4}\right)+\frac{\pi}{2}$ (D) $2\tan^{-1}2-\frac{1}{2}\log_e\left(\frac{5}{4}\right)-\frac{\pi}{2}$

Open solution

Q11. JEE Main 2026 (Online) 4th April Morning Shift

Let f be a real polynomial of degree n such that $f(x) = f'(x)f''(x)$, for all $x \in \mathbb{R}$. If $f(0)=0$, then $36\left(f'(2)+f''(2)+\int_0^2 f(x)dx\right)$ is equal to:

(A) 42 (B) 46 (C) 56 (D) 66

Open solution

Q12. JEE Main 2026 (Online) 4th April Morning Shift

Let $\int_{-2}^{2}(|\sin x|+[x\sin x])dx = 2(3-\cos 2)+\beta$, where $[\cdot]$ is the greatest integer function. Then $\beta \sin\left(\frac{\beta}{2}\right)$ equals:

(A) 1 (B) 2 (C) 4 (D) 8

Open solution

Q13. JEE Main 2026 (Online) 2nd April Evening Shift

The value of $\int_0^{20\pi}(\sin^4 x+\cos^4 x)dx$ is equal to:

(A) $\frac{15\pi}{2}$ (B) 25π (C) 15π (D) $\frac{25\pi}{2}$

Open solution

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Q14. JEE Main 2026 (Online) 2nd April Morning Shift

Let [.] denote the greatest integer function. Then the value of $\int_0^3 \left(\frac{e^x+e^{-x}}{[x]!}\right)dx$ is :

(A) $e^2+e^3-\frac{1}{e^2}-\frac{1}{e^3}$ (B) $\frac{1}{2}\left(e^2+e^3-\frac{1}{e^2}-\frac{1}{e^3}\right)$

(C) $e^2+e^3-\frac{1}{2e^2}-\frac{1}{2e^3}$ (D) $\frac{1}{2}(e^2+e^3)-\frac{1}{e^2}-\frac{1}{e^3}$

Open solution

Q15. JEE Main 2026 (Online) 28th January Evening Shift

Let [.] denote the greatest integer function. Then

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{12(3+[x])}{3+[\sin x]+[\cos x]}\right)dx$

is equal to :

(A) $12\pi+5$ (B) $11\pi+2$ (C) $15\pi+4$ (D) $13\pi+1$

Open solution

Q16. JEE Main 2026 (Online) 28th January Morning Shift

Let f be a polynomial function such that $f\left(x^2+1\right)=x^4+5x^2+2$, for all $x\in\mathbb{R}$.

Then $\int_0^3 f(x)dx$ is equal to

(A) $\frac{33}{2}$ (B) $\frac{5}{3}$ (C) $\frac{27}{2}$ (D) $\frac{41}{3}$

Open solution

Q17. JEE Main 2026 (Online) 23rd January Morning Shift

The value of the integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1+\sqrt[3]{\tan 2x}}$ is :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{18}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Open solution

Q18. JEE Main 2026 (Online) 22nd January Morning Shift

The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1}{[x]+4}\right)dx$, where $[\cdot]$ denotes the greatest integer function, is

(A) $\frac{1}{60}(21\pi-1)$ (B) $\frac{1}{60}(\pi-7)$ (C) $\frac{7}{60}(\pi-3)$ (D) $\frac{7}{60}(3\pi-1)$

Open solution

Q19. JEE Main 2026 (Online) 22nd January Morning Shift

Let $f:[1,\infty)\to\mathbb{R}$ be a differentiable function. If $6\int_1^x f(t)dt=3xf(x)+x^3-4$ for all $x\geq 1$, then the value of $f(2)-f(3)$ is :

(A) 4 (B) 3 (C) -4 (D) -3

Open solution

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Q20. JEE Main 2026 (Online) 21st January Morning Shift

The value of $\int_{-\pi/6}^{\pi/6} \left(\frac{\pi+4x^{11}}{1-\sin(|x|+\pi/6)}\right)dx$ is equal to:

(A) 8π (B) 4π (C) 2π (D) 6π

Open solution

Q21. JEE Main 2025 (Online) 8th April Evening Shift

The integral $\int_{-1}^{\frac{3}{2}} \left(|\pi^2 x \sin(\pi x)|\right)dx$ is equal to:

(A) $2+3\pi$ (B) $4+\pi$ (C) $1+3\pi$ (D) $3+2\pi$

Open solution

Q22. JEE Main 2025 (Online) 8th April Evening Shift

Let *f(x)* be a positive function and $I_1 = \int_{-\frac{1}{2}}^{1} 2x\, f(2x(1-2x))\, dx$ and $I_2 = \int_{-1}^{2} f(x(1-x))\, dx$. Then the value of $\frac{I_2}{I_1}$ is equal to _____

(A) 12 (B) 9 (C) 6 (D) 4

Open solution

Q23. JEE Main 2025 (Online) 7th April Morning Shift

The integral $\int_0^{\pi} \frac{(x+3)\sin x}{1+3\cos^2 x}dx$ is equal to

(A) $\frac{\pi}{\sqrt{3}}(\pi+1)$ (B) $\frac{\pi}{3\sqrt{3}}(\pi+6)$ (C) $\frac{\pi}{\sqrt{3}}(\pi+2)$ (D) $\frac{\pi}{2\sqrt{3}}(\pi+4)$

Open solution

Q24. JEE Main 2025 (Online) 4th April Evening Shift

Let $f(x)+2f\left(\frac{1}{x}\right)=x^2+5$ and $2g(x)-3g\left(\frac{1}{2}\right)=x, x>0$. If $\alpha=\int_1^2 f(x)\mathrm{d}x$, and $\beta=\int_1^2 g(x)\mathrm{d}x$, then the value of $9\alpha+\beta$ is :

(A) 0 (B) 10 (C) 1 (D) 11

Open solution

Q25. JEE Main 2025 (Online) 4th April Morning Shift

The value of $\int_{-1}^{1} \frac{(1+\sqrt{|x|-x})e^x+(\sqrt{|x|-x})e^{-x}}{e^x+e^{-x}}dx$ is equal to

(A) $1+\frac{2\sqrt{2}}{3}$ (B) $1-\frac{2\sqrt{2}}{3}$ (C) $2+\frac{2\sqrt{2}}{3}$ (D) $3-\frac{2\sqrt{2}}{3}$

Open solution

Q26. JEE Main 2025 (Online) 3rd April Evening Shift

The integral $\int_0^{\pi} \frac{8xdx}{4\cos^2 x+\sin^2 x}$ is equal to

(A) $2\pi^2$ (B) $4\pi^2$ (C) π^2 (D) $\frac{3\pi^2}{2}$

Open solution

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Q27. JEE Main 2025 (Online) 3rd April Morning Shift

Let the domain of the function $f(x) = \log_2 \log_4 \log_6 \left(3 + 4x - x^2\right)$ be (a, b). If $\int_0^{b-a} [x^2]\,dx = p - \sqrt{q} - \sqrt{r}, p, q, r \in \mathbb{N}, \gcd(p, q, r) = 1$, where $[\cdot]$ is the greatest integer function, then $p + q + r$ is equal to

(A) 10 (B) 11 (C) 9 (D) 8

Open solution

Q28. JEE Main 2025 (Online) 2nd April Evening Shift

Let $f : [1, \infty) \to [2, \infty)$ be a differentiable function. If $10 \int_1^x f(\mathrm{t})\mathrm{dt} = 5xf(x) - x^5 - 9$ for all $x \geqslant 1$, then the value of $f(3)$ is :

(A) 22 (B) 26 (C) 32 (D) 18

Open solution

Q29. JEE Main 2025 (Online) 2nd April Evening Shift

Let (a, b) be the point of intersection of the curve $x^2 = 2y$ and the straight line $y - 2x - 6 = 0$ in the second quadrant. Then the integral $\mathrm{I} = \int_{\mathrm{a}}^{\mathrm{b}} \frac{9x^2}{1+5^x}\,\mathrm{d}x$ is equal to :

(A) 27 (B) 18 (C) 24 (D) 21

Open solution

Q30. JEE Main 2025 (Online) 2nd April Evening Shift

$4\int_0^1 \left(\frac{1}{\sqrt{3+x^2}+\sqrt{1+x^2}}\right)dx - 3\log_e(\sqrt{3})$ is equal to :

(A) $2 - \sqrt{2} - \log_e(1 + \sqrt{2})$ (B) $2 + \sqrt{2} + \log_e(1 + \sqrt{2})$

(C) $2 + \sqrt{2} - \log_e(1 + \sqrt{2})$ (D) $2 - \sqrt{2} + \log_e(1 + \sqrt{2})$

Open solution

Q31. JEE Main 2025 (Online) 29th January Evening Shift

Let $f(x) = \int_0^x \mathrm{t}\left(\mathrm{t}^2 - 9\mathrm{t} + 20\right)\mathrm{dt}, 1 \leq x \leq 5$. If the range of f is $[\alpha, \beta]$, then $4(\alpha + \beta)$ equals :

(A) 253 (B) 157 (C) 154 (D) 125

Open solution

Q32. JEE Main 2025 (Online) 29th January Morning Shift

The integral $80\int_0^{\frac{\pi}{4}} \left(\frac{\sin\theta+\cos\theta}{9+16\sin 2\theta}\right)d\theta$ is equal to :

(A) $3\log 4$ (B) $4\log 3$ (C) $6\log\frac{4}{3}$ (D) $2\log 3$

Open solution

Q33. JEE Main 2025 (Online) 28th January Evening Shift

Let $\mathrm{f} : \mathrm{R} \to \mathrm{R}$ be a twice differentiable function such that $f(2) = 1$. If $\mathrm{F(x)} = \mathrm{x}f(\mathrm{x})$ for all $\mathrm{x} \in \mathrm{R}$, $\int_0^2 xF'(x)dx = 6$ and $\int_0^2 x^2F''(x)dx = 40$, then $F'(2) + \int_0^2 F(x)dx$ is equal to :

(A) 13 (B) 11 (C) 9 (D) 15

Open solution

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Q34. JEE Main 2025 (Online) 28th January Evening Shift

Let f be a real valued continuous function defined on the positive real axis such that $g(x) = \int_0^x tf(t)dt$. If $g\left(x^3\right) = x^6 + x^7$, then value of $\sum_{r=1}^{15} f\left(r^3\right)$ is :

(A) 270 (B) 340 (C) 310 (D) 320

Open solution

Q35. JEE Main 2025 (Online) 28th January Morning Shift

If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2\cos^2 x}{(1+e^x)}\mathrm{d}x = \pi\left(\alpha\pi^2 + \beta\right), \alpha, \beta \in \mathbb{Z}$, then $(\alpha+\beta)^2$ equals

(A) 196 (B) 100 (C) 64 (D) 144

Open solution

Q36. JEE Main 2025 (Online) 24th January Morning Shift

If $I(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}dx, m,n>0$, then $I(9,14)+I(10,13)$ is

(A) $I(9,1)$ (B) $I(1,13)$ (C) $\mathrm{I}(19,27)$ (D) $\mathrm{I}(9,13)$

Open solution

Q37. JEE Main 2025 (Online) 23rd January Evening Shift

If $\mathrm{I} = \int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{3}{2}} x}{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}\,\mathrm{d}x$, then $\int_0^{2\mathrm{I}} \frac{x\sin x\cos x}{\sin^4 x+\cos^4 x}\,\mathrm{d}x$ equals :

(A) $\frac{\pi^2}{12}$ (B) $\frac{\pi^2}{4}$ (C) $\frac{\pi^2}{16}$ (D) $\frac{\pi^2}{8}$

Open solution

Q38. JEE Main 2025 (Online) 23rd January Morning Shift

The value of $\int_{e^2}^{e^4} \frac{1}{x}\left(\frac{e^{\left((\log_e x)^2+1\right)^{-1}}}{e^{\left((\log_e x)^2+1\right)^{-1}}+e^{\left((6-\log_e x)^2+1\right)^{-1}}}\right)dx$ is

(A) 1 (B) $\log_e 2$ (C) e^2 (D) 2

Open solution

Q39. JEE Main 2025 (Online) 22nd January Morning Shift

Let for $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$, $\mathrm{I}_1 = \int_0^{\pi/4} f(x)\mathrm{d}x$ and $\mathrm{I}_2 = \int_0^{\pi/4} xf(x)\mathrm{d}x$.
Then $7\mathrm{I}_1 + 12\mathrm{I}_2$ is equal to :

(A) 2π (B) 1 (C) π (D) 2

Open solution

Q40. JEE Main 2024 (Online) 9th April Evening Shift

The integral $\int_{1/4}^{3/4} \cos\left(2\cot^{-1}\sqrt{\frac{1-x}{1+x}}\right)dx$ is equal to

(A) $-1/2$ (B) $-1/4$ (C) 1/4 (D) 1/2

Open solution

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Q41. JEE Main 2024 (Online) 9th April Evening Shift

$\lim\limits_{x\to\frac{\pi}{2}}\left(\frac{\int_{x^3}^{(\pi/2)^3}\left(\sin\left(2t^{1/3}\right)+\cos\left(t^{1/3}\right)\right)dt}{\left(x-\frac{\pi}{2}\right)^2}\right)$ is equal to

(A) $\frac{3\pi^2}{2}$ (B) $\frac{9\pi^2}{8}$ (C) $\frac{5\pi^2}{9}$ (D) $\frac{11\pi^2}{10}$

Open solution

Q42. JEE Main 2024 (Online) 9th April Evening Shift

The value of the integral $\int_{-1}^{2}\log_e\left(x+\sqrt{x^2+1}\right)dx$ is

(A) $\sqrt{5}-\sqrt{2}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$ (B) $\sqrt{2}-\sqrt{5}+\log_e\left(\frac{7+4\sqrt{5}}{1+\sqrt{2}}\right)$

(C) $\sqrt{5}-\sqrt{2}+\log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$ (D) $\sqrt{2}-\sqrt{5}+\log_e\left(\frac{9+4\sqrt{5}}{1+\sqrt{2}}\right)$

Open solution

Q43. JEE Main 2024 (Online) 8th April Evening Shift

Let $\int_{\alpha}^{\log_e 4}\frac{dx}{\sqrt{e^x-1}}=\frac{\pi}{6}$. Then e^{α} and $e^{-\alpha}$ are the roots of the equation :

(A) $2x^2-5x+2=0$ (B) $x^2-2x-8=0$ (C) $2x^2-5x-2=0$ (D) $x^2+2x-8=0$

Open solution

Q44. JEE Main 2024 (Online) 8th April Morning Shift

The value of $k\in\mathbb{N}$ for which the integral $I_n=\int_0^1\left(1-x^k\right)^n dx, n\in\mathbb{N}$, satisfies $147I_{20}=148I_{21}$ is

(A) 8 (B) 14 (C) 7 (D) 10

Open solution

Q45. JEE Main 2024 (Online) 6th April Morning Shift

$\int_0^{\pi/4}\frac{\cos^2 x\sin^2 x}{\left(\cos^3 x+\sin^3 x\right)^2}dx$ is equal to

(A) 1/9 (B) 1/6 (C) 1/3 (D) 1/12

Open solution

Q46. JEE Main 2024 (Online) 5th April Evening Shift

Let $\beta(\mathrm{m},\mathrm{n})=\int_0^1 x^{\mathrm{m}-1}(1-x)^{\mathrm{n}-1}\,dx,\ \mathrm{m},\mathrm{n}>0$. If $\int_0^1\left(1-x^{10}\right)^{20}dx=\mathrm{a}\times\beta(\mathrm{b},\mathrm{c})$, then $100(\mathrm{a}+\mathrm{b}+\mathrm{c})$ equals _____.

(A) 2012 (B) 1021 (C) 1120 (D) 2120

Open solution

Q47. JEE Main 2024 (Online) 5th April Morning Shift

The integral $\int_0^{\pi/4}\frac{136\sin x}{3\sin x+5\cos x}dx$ is equal to :

(A) $3\pi-50\log_e 2+20\log_e 5$ (B) $3\pi-25\log_e 2+10\log_e 5$

(C) $3\pi-10\log_e(2\sqrt{2})+10\log_e 5$ (D) $3\pi-30\log_e 2+20\log_e 5$

Open solution

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Q48. JEE Main 2024 (Online) 5th April Morning Shift

The value of $\int_{-\pi}^{\pi} \frac{2y(1+\sin y)}{1+\cos^2 y}dy$ is :

(A) $\frac{\pi}{2}$ (B) π^2 (C) $\frac{\pi^2}{2}$ (D) $2\pi^2$

Open solution

Q49. JEE Main 2024 (Online) 4th April Evening Shift

Let $f(x) = \int_0^x \left(t + \sin\left(1 - e^t\right)\right)dt, x \in \mathbb{R}$. Then, $\lim_{x\to 0} \frac{f(x)}{x^3}$ is equal to

(A) $\frac{1}{6}$ (B) $-\frac{1}{6}$ (C) $\frac{2}{3}$ (D) $-\frac{2}{3}$

Open solution

Q50. JEE Main 2024 (Online) 4th April Evening Shift

If the value of the integral $\int_{-1}^{1} \frac{\cos \alpha x}{1+3^x}dx$ is $\frac{2}{\pi}$, then a value of α is

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Open solution

Q51. JEE Main 2024 (Online) 4th April Morning Shift

Let $f(x) = \begin{cases} -2, & -2 \le x \le 0 \\ x-2, & 0 < x \le 2 \end{cases}$ and $\mathrm{h}(x) = f(|x|) + |f(x)|$. Then $\int_{-2}^{2} \mathrm{h}(x)\mathrm{d}x$ is equal to:

(A) 2 (B) 6 (C) 4 (D) 1

Open solution

Q52. JEE Main 2024 (Online) 1st February Evening Shift

If $\int_0^{\frac{\pi}{3}} \cos^4 x \, \mathrm{d}x = \mathrm{a}\pi + \mathrm{b}\sqrt{3}$, where a and b are rational numbers, then $9\mathrm{a} + 8\mathrm{b}$ is equal to :

(A) 2 (B) 1 (C) 3 (D) $\frac{3}{2}$

Open solution

Q53. JEE Main 2024 (Online) 1st February Evening Shift

The value of $\int_0^1 \left(2x^3 - 3x^2 - x + 1\right)^{\frac{1}{3}} \mathrm{d}x$ is equal to :

(A) -1 (B) 2 (C) 0 (D) 1

Open solution

Q54. JEE Main 2024 (Online) 1st February Morning Shift

The value of the integral $\int_0^{\pi/4} \frac{x \, \mathrm{d}x}{\sin^4(2x)+\cos^4(2x)}$ equals :

(A) $\frac{\sqrt{2}\pi^2}{8}$ (B) $\frac{\sqrt{2}\pi^2}{16}$ (C) $\frac{\sqrt{2}\pi^2}{32}$ (D) $\frac{\sqrt{2}\pi^2}{64}$

Open solution

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Q55. JEE Main 2024 (Online) 31st January Evening Shift

Let $f, g : (0, \infty) \to \mathbb{R}$ be two functions defined by $f(x) = \int_{-x}^{x} (|t| - t^2)e^{-t^2}dt$ and $g(x) = \int_{0}^{x^2} t^{1/2}e^{-t}dt$. Then, the value of $9\left(f\left(\sqrt{\log_e 9}\right) + g\left(\sqrt{\log_e 9}\right)\right)$ is equal to :

(A) 10 (B) 9 (C) 8 (D) 6

Open solution

Q56. JEE Main 2024 (Online) 30th January Evening Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \frac{x}{(1+x^4)^{1/4}}$, and $g(x) = f(f(f(f(x))))$. Then, $18\int_0^{\sqrt{2\sqrt{5}}} x^2 g(x)dx$ is equal to

(A) 36 (B) 33 (C) 39 (D) 42

Open solution

Q57. JEE Main 2024 (Online) 30th January Evening Shift

Let $y = f(x)$ be a thrice differentiable function in $(-5, 5)$. Let the tangents to the curve $y = f(x)$ at $(1, f(1))$ and $(3, f(3))$ make angles $\pi/6$ and $\pi/4$, respectively with positive x-axis. If $27\int_1^3 \left((f'(t))^2 + 1\right) f''(t)dt = \alpha + \beta\sqrt{3}$ where α, β are integers, then the value of $\alpha + \beta$ equals

(A) 26 (B) -16 (C) 36 (D) -14

Open solution

Q58. JEE Main 2024 (Online) 30th January Evening Shift

Let a and b be real constants such that the function f defined by $f(x) = \begin{cases} x^2 + 3x + a & , x \le 1 \\ bx + 2 & , x > 1 \end{cases}$ be differentiable on $\mathbb{R}$. Then, the value of $\int_{-2}^{2} f(x)dx$ equals

(A) 21 (B) 19/6 (C) 17 (D) 15/6

Open solution

Q59. JEE Main 2024 (Online) 30th January Evening Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = ae^{2x} + be^x + cx$. If $f(0) = -1$, $f'(\log_e 2) = 21$ and $\int_0^{\log_e 4}(f(x) - cx)dx = \frac{39}{2}$, then the value of $|a + b + c|$ equals

(A) 16 (B) 12 (C) 8 (D) 10

Open solution

Q60. JEE Main 2024 (Online) 30th January Morning Shift

The value of $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n^3}{(n^2+k^2)(n^2+3k^2)}$ is :

(A) $\frac{\pi}{8(2\sqrt{3}+3)}$ (B) $\frac{(2\sqrt{3}+3)\pi}{24}$ (C) $\frac{13\pi}{8(4\sqrt{3}+3)}$ (D) $\frac{13(2\sqrt{3}-3)\pi}{8}$

Open solution

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Q61. JEE Main 2024 (Online) 30th January Morning Shift

Let $f:\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbf{R}$ be a differentiable function such that $f(0)=\frac{1}{2}$. If the $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{e^{x^2}-1}=\alpha$, then $8 \alpha^2$ is equal to :

(A) 4 (B) 2 (C) 1 (D) 16

Open solution

Q62. JEE Main 2024 (Online) 29th January Morning Shift

$\lim_{x \rightarrow \frac{\pi}{2}}\left(\frac{1}{\left(x-\frac{\pi}{2}\right)^2} \int_{x^3}^{\left(\frac{\pi}{2}\right)^3} \cos \left(t^{\frac{1}{3}}\right) dt\right)$ is equal to

(A) $\frac{3\pi^2}{4}$ (B) $\frac{3\pi^2}{8}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{8}$

Open solution

Q63. JEE Main 2024 (Online) 29th January Morning Shift

If the value of the integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(\frac{x^2 \cos x}{1+\pi^x}+\frac{1+\sin^2 x}{1+e^{\sin x^{2023}}}\right) dx=\frac{\pi}{4}(\pi+a)-2$, then the value of a is

(A) $-\frac{3}{2}$ (B) 3 (C) $\frac{3}{2}$ (D) 2

Open solution

Q64. JEE Main 2024 (Online) 27th January Evening Shift

For $0<\mathrm{a}<1$, the value of the integral $\int_0^{\pi} \frac{dx}{1-2a \cos x+a^2}$ is :

(A) $\frac{\pi^2}{\pi+a^2}$ (B) $\frac{\pi^2}{\pi-a^2}$ (C) $\frac{\pi}{1-a^2}$ (D) $\frac{\pi}{1+a^2}$

Open solution

Q65. JEE Main 2024 (Online) 27th January Morning Shift

If $\int_0^1 \frac{1}{\sqrt{3+x}+\sqrt{1+x}} dx=a+b \sqrt{2}+c \sqrt{3}$, where a, b, c are rational numbers, then $2a+3b-4c$ is equal to :

(A) 10 (B) 7 (C) 4 (D) 8

Open solution

Q66. JEE Main 2024 (Online) 27th January Morning Shift

If (a, b) be the orthocentre of the triangle whose vertices are $(1,2),(2,3)$ and $(3,1)$, and $I_1=\int_a^b x \sin \left(4x-x^2\right) dx, I_2=\int_a^b \sin \left(4x-x^2\right) dx$, then $36 \frac{I_1}{I_2}$ is equal to :

(A) 80 (B) 72 (C) 66 (D) 88

Open solution

Q67. JEE Main 2023 (Online) 15th April Morning Shift

If $\int_0^1 \frac{1}{(5+2x-2x^2)\left(1+e^{(2-4x)}\right)} dx=\frac{1}{\alpha} \log_e\left(\frac{\alpha+1}{\beta}\right), \alpha, \beta>0$, then $\alpha^4-\beta^4$ is equal to :

(A) -21 (B) 21 (C) 19 (D) 0

Open solution

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Q68. JEE Main 2023 (Online) 13th April Evening Shift

The value of $\frac{e^{-\frac{\pi}{4}}+\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx}{\int_0^{\frac{\pi}{4}} e^{-x}(\tan^{49}x+\tan^{51}x)\,dx}$ is

Open solution

(A) 51 (B) 50 (C) 25 (D) 49

Q69. JEE Main 2023 (Online) 13th April Morning Shift

$\int_0^{\infty} \frac{6}{e^{3x}+6e^{2x}+11e^{x}+6}dx =$

Open solution

(A) $\log_e\left(\frac{256}{81}\right)$ (B) $\log_e\left(\frac{64}{27}\right)$

(C) $\log_e\left(\frac{32}{27}\right)$ (D) $\log_e\left(\frac{512}{81}\right)$

Q70. JEE Main 2023 (Online) 11th April Evening Shift

If $f:\mathbb{R}\to\mathbb{R}$ be a continuous function satisfying $\int_0^{\frac{\pi}{2}} f(\sin 2x)\sin x\,dx+\alpha\int_0^{\frac{\pi}{4}} f(\cos 2x)\cos x\,dx=0$, then the value of α is :

Open solution

(A) $-\sqrt{3}$ (B) $\sqrt{2}$ (C) $-\sqrt{2}$ (D) $\sqrt{3}$

Q71. JEE Main 2023 (Online) 11th April Evening Shift

Let the function $f:[0,2]\to\mathbb{R}$ be defined as

$$f(x)=\begin{cases} e^{\min\{x^2,x-[x]\}}, & x\in[0,1) \\ e^{[x-\log_e x]}, & x\in[1,2] \end{cases}$$

where $[t]$ denotes the greatest integer less than or equal to t. Then the value of the integral $\int_0^2 xf(x)dx$ is :

Open solution

(A) $2e-1$ (B) $2e-\frac{1}{2}$

(C) $1+\frac{3e}{2}$ (D) $(e-1)\left(e^2+\frac{1}{2}\right)$

Q72. JEE Main 2023 (Online) 11th April Morning Shift

The value of the integral $\int_{-\log_e 2}^{\log_e 2} e^x\left(\log_e\left(e^x+\sqrt{1+e^{2x}}\right)\right)dx$ is equal to :

Open solution

(A) $\log_e\left(\frac{(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$

(B) $\log_e\left(\frac{\sqrt{2}(2+\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$

(C) $\log_e\left(\frac{2(2+\sqrt{5})}{\sqrt{1+\sqrt{5}}}\right)-\frac{\sqrt{5}}{2}$

(D) $\log_e\left(\frac{\sqrt{2}(3-\sqrt{5})^2}{\sqrt{1+\sqrt{5}}}\right)+\frac{\sqrt{5}}{2}$

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Q73. JEE Main 2023 (Online) 10th April Evening Shift

Let f be a continuous function satisfying $\int_0^{t^2} \left(f(x)+x^2\right)dx = \frac{4}{3}t^3, \forall t > 0$. Then $f\left(\frac{\pi^2}{4}\right)$ is equal to :

(A) $-\pi\left(1+\frac{\pi^3}{16}\right)$ (B) $\pi\left(1-\frac{\pi^3}{16}\right)$

(C) $-\pi^2\left(1+\frac{\pi^2}{16}\right)$ (D) $\pi^2\left(1-\frac{\pi^2}{16}\right)$

Open solution

Q74. JEE Main 2023 (Online) 6th April Evening Shift

Let $f(x)$ be a function satisfying $f(x)+f(\pi-x)=\pi^2, \forall x \in \mathbb{R}$. Then $\int_0^{\pi} f(x)\sin x dx$ is equal to :

(A) π^2 (B) $\frac{\pi^2}{2}$ (C) $2\pi^2$ (D) $\frac{\pi^2}{4}$

Open solution

Q75. JEE Main 2023 (Online) 6th April Morning Shift

Let $5f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}+3, x>0$. Then $18\int_1^2 f(x)dx$ is equal to :

(A) $10\log_e 2+6$ (B) $5\log_e 2-3$ (C) $10\log_e 2-6$ (D) $5\log_e 2+3$

Open solution

Q76. JEE Main 2023 (Online) 1st February Evening Shift

The value of the integral

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x+\frac{\pi}{4}}{2-\cos 2x}dx$ is :

(A) $\frac{\pi^2}{6\sqrt{3}}$ (B) $\frac{\pi^2}{6}$ (C) $\frac{\pi^2}{3\sqrt{3}}$ (D) $\frac{\pi^2}{12\sqrt{3}}$

Open solution

Q77. JEE Main 2023 (Online) 31st January Evening Shift

Let $\alpha > 0$. If $\int_0^{\alpha} \frac{x}{\sqrt{x+\alpha}-\sqrt{x}}\,dx = \frac{16+20\sqrt{2}}{15}$, then α is equal to :

(A) 4 (B) 2 (C) $2\sqrt{2}$ (D) $\sqrt{2}$

Open solution

Q78. JEE Main 2023 (Online) 31st January Evening Shift

If $\phi(x)=\frac{1}{\sqrt{x}}\int_{\frac{\pi}{4}}^{x}\left(4\sqrt{2}\sin t-3\phi'(t)\right)dt, x>0$,

then $\emptyset'\left(\frac{\pi}{4}\right)$ is equal to :

(A) $\frac{4}{6+\sqrt{\pi}}$ (B) $\frac{4}{6-\sqrt{\pi}}$ (C) $\frac{8}{\sqrt{\pi}}$ (D) $\frac{8}{6+\sqrt{\pi}}$

Open solution

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Q79. JEE Main 2023 (Online) 31st January Morning Shift

Let $\alpha \in (0,1)$ and $\beta = \log_e(1-\alpha)$. Let $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots + \frac{x^n}{n}, x \in (0,1)$. Then the integral $\int_0^{\alpha} \frac{t^{50}}{1-t} dt$ is equal to

Open solution

(A) $-(\beta + P_{50}(\alpha))$

(B) $\beta - P_{50}(\alpha)$

(C) $P_{50}(\alpha) - \beta$

(D) $\beta + P_{50}(\alpha)$

Q80. JEE Main 2023 (Online) 31st January Morning Shift

The value of $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{(2+3\sin x)}{\sin x(1+\cos x)} dx$ is equal to :

Open solution

(A) $\frac{10}{3} - \sqrt{3} + \log_e \sqrt{3}$

(B) $\frac{7}{2} - \sqrt{3} - \log_e \sqrt{3}$

(C) $\frac{10}{3} - \sqrt{3} - \log_e \sqrt{3}$

(D) $-2 + 3\sqrt{3} + \log_e \sqrt{3}$

Q81. JEE Main 2023 (Online) 30th January Morning Shift

If [t] denotes the greatest integer $\leq$ t, then the value of $\frac{3(e-1)}{e} \int_1^2 x^2 e^{[x]+[x^3]} dx$ is :

Open solution

(A) $e^8 - e$

(B) $e^7 - 1$

(C) $e^9 - e$

(D) $e^8 - 1$

Q82. JEE Main 2023 (Online) 29th January Evening Shift

The value of the integral $\int_1^2 \left(\frac{t^4+1}{t^6+1}\right) dt$ is

Open solution

(A) $\tan^{-1}\frac{1}{2} - \frac{1}{3}\tan^{-1}8 + \frac{\pi}{3}$

(B) $\tan^{-1}2 - \frac{1}{3}\tan^{-1}8 + \frac{\pi}{3}$

(C) $\tan^{-1}2 + \frac{1}{3}\tan^{-1}8 - \frac{\pi}{3}$

(D) $\tan^{-1}\frac{1}{2} + \frac{1}{3}\tan^{-1}8 - \frac{\pi}{3}$

Q83. JEE Main 2023 (Online) 29th January Evening Shift

The value of the integral $\int_{1/2}^{2} \frac{\tan^{-1}x}{x} dx$ is equal to :

Open solution

(A) $\frac{\pi}{2}\log_e 2$

(B) $\frac{\pi}{4}\log_e 2$

(C) $\frac{1}{2}\log_e 2$

(D) $\pi\log_e 2$

Q84. JEE Main 2023 (Online) 29th January Morning Shift

Let $f(x) = x + \frac{a}{\pi^2-4}\sin x + \frac{b}{\pi^2-4}\cos x, x \in R$ be a function which satisfies $f(x) = x + \int_0^{\pi/2} \sin(x+y) f(y) dy$. then $(a+b)$ is equal to

Open solution

(A) $-2\pi(\pi+2)$

(B) $-\pi(\pi-2)$

(C) $-\pi(\pi+2)$

(D) $-2\pi(\pi-2)$

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Q85. JEE Main 2023 (Online) 29th January Morning Shift

Let $[x]$ denote the greatest integer $\leq x$. Consider the function $f(x) = \max\{x^2, 1 + [x]\}$. Then the value of the integral $\int_0^2 f(x)dx$ is

(A) $\frac{5+4\sqrt{2}}{3}$ (B) $\frac{4+5\sqrt{2}}{3}$ (C) $\frac{8+4\sqrt{2}}{3}$ (D) $\frac{1+5\sqrt{2}}{3}$

Open solution

Q86. JEE Main 2023 (Online) 25th January Evening Shift

The integral $16\int_1^2 \frac{dx}{x^3(x^2+2)^2}$ is equal to

(A) $\frac{11}{12} + \log_e 4$ (B) $\frac{11}{6} + \log_e 4$ (C) $\frac{11}{12} - \log_e 4$ (D) $\frac{11}{6} - \log_e 4$

Open solution

Q87. JEE Main 2023 (Online) 25th January Morning Shift

The minimum value of the function $f(x) = \int_0^2 e^{|x-t|}dt$ is :

(A) 2 (B) $2(e-1)$ (C) $e(e-1)$ (D) $2e-1$

Open solution

Q88. JEE Main 2023 (Online) 24th January Evening Shift

$\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{9-4x^2}}dx$ is equal to :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) 2π

Open solution

Q89. JEE Main 2022 (Online) 30th July Evening Shift

If $y(x) = \tan x - \int_0^{2x} (x^2 - xt + 1)\sin t\, dt$, then at $x = \frac{\pi}{3}$, the value of $2y''' - y''$ is equal to

(A) 158 (B) $159 - 4\sqrt{3}$ (C) 126 (D) $127 - 4\sqrt{3}$

Open solution

Q90. JEE Main 2022 (Online) 30th July Evening Shift

Let $A_k = \int_{k\pi}^{(k+1)\pi} e^{-x}\sin x\, dx$, $k = 0, 1, 2, \ldots, 20$. Then $\sum_{k=0}^{20} |A_k|$ is equal to

(A) $\left(\frac{e^\pi+1}{e^\pi-1}\right)\left(1-e^{-20\pi}\right)$ (B) $\frac{1}{2}\left(\frac{e^\pi+1}{e^\pi-1}\right)\left(1-e^{-21\pi}\right)$

(C) $\left(\frac{e^\pi+1}{e^\pi-1}\right)\left(1-e^{-21\pi}\right)$ (D) $\frac{1}{2}\left(\frac{e^\pi+1}{e^\pi-1}\right)\left(1-e^{-20\pi}\right)$

Open solution

Q91. JEE Main 2022 (Online) 29th July Evening Shift

If $[t]$ denotes the greatest integer $\leq t$, then the value of $\int_0^1 \left[2x - \left|3x^2 - 5x + 2\right| + 1\right]dx$ is :

(A) $\frac{\sqrt{37}+\sqrt{13}-4}{6}$ (B) $\frac{\sqrt{37}-\sqrt{13}-4}{6}$ (C) $\frac{-\sqrt{37}-\sqrt{13}+4}{6}$ (D) $\frac{-\sqrt{37}+\sqrt{13}+4}{6}$

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Q92. JEE Main 2022 (Online) 29th July Morning Shift

The integral $\int_0^{\frac{\pi}{2}} \frac{1}{3+2\sin x+\cos x}\,dx$ is equal to :

(A) $\tan^{-1}(2)$

(B) $\tan^{-1}(2)-\frac{\pi}{4}$

(C) $\frac{1}{2}\tan^{-1}(2)-\frac{\pi}{8}$

(D) $\frac{1}{2}$

Open solution

Q93. JEE Main 2022 (Online) 29th July Morning Shift

If $f(\alpha)=\int_1^{\alpha}\frac{\log_{10}t}{1+t}dt, \alpha>0$, then $f\left(e^3\right)+f\left(e^{-3}\right)$ is equal to :

(A) 9

(B) $\frac{9}{2}$

(C) $\frac{9}{\log_e(10)}$

(D) $\frac{9}{2\log_e(10)}$

Open solution

Q94. JEE Main 2022 (Online) 28th July Evening Shift

Let $I_n(x)=\int_0^x \frac{1}{(t^2+5)^n}dt, n=1,2,3,\ldots.$ Then :

(A) $50I_6-9I_5=xI_5'$

(B) $50I_6-11I_5=xI_5'$

(C) $50I_6-9I_5=I_5'$

(D) $50I_6-11I_5=I_5'$

Open solution

Q95. JEE Main 2022 (Online) 28th July Morning Shift

The minimum value of the twice differentiable function $f(x)=\int_0^x e^{x-t}f'(t)dt-\left(x^2-x+1\right)e^x$, $x\in\mathbf{R}$, is :

(A) $-\frac{2}{\sqrt{e}}$

(B) $-2\sqrt{e}$

(C) $-\sqrt{e}$

(D) $\frac{2}{\sqrt{e}}$

Open solution

Q96. JEE Main 2022 (Online) 27th July Evening Shift

Let $f(x)=2+|x|-|x-1|+|x+1|, x\in\mathbf{R}$.

Consider

(S1) : $f'\left(-\frac{3}{2}\right)+f'\left(-\frac{1}{2}\right)+f'\left(\frac{1}{2}\right)+f'\left(\frac{3}{2}\right)=2$

(S2) : $\int_{-2}^{2} f(x)dx=12$

Then,

(A) both (S1) and (S2) are correct

(B) both (S1) and (S2) are wrong

(C) only (S1) is correct

(D) only (S2) is correct

Open solution

Q97. JEE Main 2022 (Online) 27th July Evening Shift

$\int_0^2 \left(\left|2x^2-3x\right|+\left[x-\frac{1}{2}\right]\right)dx$, where [t] is the greatest integer function, is equal to :

(A) $\frac{7}{6}$

(B) $\frac{19}{12}$

(C) $\frac{31}{12}$

(D) $\frac{3}{2}$

Open solution

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Q98. JEE Main 2022 (Online) 27th July Morning Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined as

$f(x) = a \sin\left(\frac{\pi[x]}{2}\right) + [2 - x], a \in \mathbb{R}$ where $[t]$ is the greatest integer less than or equal to t. If $\lim\limits_{x \to -1} f(x)$ exists, then the value of $\int\limits_0^4 f(x)dx$ is equal to

(A) -1 (B) -2 (C) 1 (D) 2

Open solution

Q99. JEE Main 2022 (Online) 27th July Morning Shift

Let $I = \int_{\pi/4}^{\pi/3} \left(\frac{8\sin x - \sin 2x}{x}\right)dx$. Then

(A) $\frac{\pi}{2} < I < \frac{3\pi}{4}$ (B) $\frac{\pi}{5} < I < \frac{5\pi}{12}$

(C) $\frac{5\pi}{12} < I < \frac{\sqrt{2}}{3}\pi$ (D) $\frac{3\pi}{4} < I < \pi$

Open solution

Q100. JEE Main 2022 (Online) 27th July Morning Shift

Let a function $f : \mathbb{R} \to \mathbb{R}$ be defined as :

$$f(x) = \begin{cases} \int\limits_0^x (5 - |t - 3|)dt, & x > 4 \\ x^2 + bx & , x \leq 4 \end{cases}$$

where $b \in \mathbb{R}$. If f is continuous at $x = 4$, then which of the following statements is NOT true?

(A) f is not differentiable at $x = 4$

(B) $f'(3) + f'(5) = \frac{35}{4}$

(C) f is increasing in $\left(-\infty, \frac{1}{8}\right) \cup (8, \infty)$

(D) f has a local minima at $x = \frac{1}{8}$

Open solution

Q101. JEE Main 2022 (Online) 26th July Evening Shift

$\int\limits_0^{20\pi} (|\sin x| + |\cos x|)^2 dx$ is equal to

(A) $10(\pi + 4)$ (B) $10(\pi + 2)$ (C) $20(\pi - 2)$ (D) $20(\pi + 2)$

Open solution

Q102. JEE Main 2022 (Online) 25th July Evening Shift

Let $[t]$ denote the greatest integer less than or equal to t. Then the value of the integral $\int_{-3}^{101} \left([\sin(\pi x)] + e^{[\cos(2\pi x)]}\right)dx$ is equal to

(A) $\frac{52(1-e)}{e}$ (B) $\frac{52}{e}$ (C) $\frac{52(2+e)}{e}$ (D) $\frac{104}{e}$

Open solution

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Q103. JEE Main 2022 (Online) 25th July Morning Shift

For any real number x, let $[x]$ denote the largest integer less than equal to x. Let f be a real valued function defined on the interval $[-10, 10]$ by $f(x) = \begin{cases} x - [x], & \text{if } [x] \text{ is odd} \\ 1 + [x] - x, & \text{if } [x] \text{ is even} \end{cases}$. Then the value of $\frac{\pi^2}{10} \int_{-10}^{10} f(x) \cos \pi x \, dx$ is :

(A) 4 (B) 2 (C) 1 (D) 0

Open solution

Q104. JEE Main 2022 (Online) 29th June Evening Shift

Let f be a real valued continuous function on [0, 1] and $f(x) = x + \int_0^1 (x - t) f(t) dt$.

Then, which of the following points (x, y) lies on the curve y = f(x) ?

(A) (2, 4) (B) (1, 2) (C) (4, 17) (D) (6, 8)

Open solution

Q105. JEE Main 2022 (Online) 29th June Evening Shift

If $\int_0^2 \left(\sqrt{2x} - \sqrt{2x - x^2}\right) dx = \int_0^1 \left(1 - \sqrt{1 - y^2} - \frac{y^2}{2}\right) dy + \int_1^2 \left(2 - \frac{y^2}{2}\right) dy + I$, then I equals

(A) $\int_0^1 \left(1 + \sqrt{1 - y^2}\right) dy$ (B) $\int_0^1 \left(\frac{y^2}{2} - \sqrt{1 - y^2} + 1\right) dy$

(C) $\int_0^1 \left(1 - \sqrt{1 - y^2}\right) dy$ (D) $\int_0^1 \left(\frac{y^2}{2} + \sqrt{1 - y^2} + 1\right) dy$

Open solution

Q106. JEE Main 2022 (Online) 29th June Morning Shift

Let $f : R \to R$ be a function defined by :

$$f(x) = \begin{cases} \max\{t^3 - 3t : t \le x\} & ; \quad x \le 2 \\ x^2 + 2x - 6 & ; \quad 2 < x < 3 \\ [x - 3] + 9 & ; \quad 3 \le x \le 5 \\ 2x + 1 & ; \quad x > 5 \end{cases}$$

where [t] is the greatest integer less than or equal to t. Let m be the number of points where f is not differentiable and $I = \int_{-2}^{2} f(x) \, dx$. Then the ordered pair (m, I) is equal to :

(A) $\left(3, \frac{27}{4}\right)$ (B) $\left(3, \frac{23}{4}\right)$

(C) $\left(4, \frac{27}{4}\right)$ (D) $\left(4, \frac{23}{4}\right)$

Open solution

Q107. JEE Main 2022 (Online) 29th June Morning Shift

$\int_0^5 \cos\left(\pi\left(x - \left[\frac{x}{2}\right]\right)\right) dx$,

where [t] denotes greatest integer less than or equal to t, is equal to:

(A) -3 (B) -2 (C) 2 (D) 0

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Q108. JEE Main 2022 (Online) 28th June Evening Shift

Let f : R $\rightarrow$ R be a differentiable function such that $f\left(\frac{\pi}{4}\right)=\sqrt{2}$, $f\left(\frac{\pi}{2}\right)=0$ and $f'\left(\frac{\pi}{2}\right)=1$ and let $g(x)=\int_x^{\pi/4}\left(f'(t)\sec t+\tan t\sec t\, f(t)\right)dt$ for $x\in\left[\frac{\pi}{4},\frac{\pi}{2}\right)$. Then $\lim\limits_{x\to\left(\frac{\pi}{2}\right)^-} g(x)$ is equal to :

Open solution

(A) 2 (B) 3 (C) 4 (D) -3

Q109. JEE Main 2022 (Online) 28th June Evening Shift

Let f : R $\rightarrow$ R be a continuous function satisfying f(x) + f(x + k) = n, for all x $\in$ R where k > 0 and n is a positive integer. If $I_1=\int\limits_0^{4nk} f(x)dx$ and $I_2=\int\limits_{-k}^{3k} f(x)dx$, then :

Open solution

(A) $I_1+2I_2=4nk$ (B) $I_1+2I_2=2nk$ (C) $I_1+nI_2=4n^2k$ (D) $I_1+nI_2=6n^2k$

Q110. JEE Main 2022 (Online) 28th June Morning Shift

Let [t] denote the greatest integer less than or equal to t. Then, the value of the integral $\int\limits_0^1[-8x^2+6x-1]dx$ is equal to :

Open solution

(A) -1 (B) $\frac{-5}{4}$ (C) $\frac{\sqrt{17}-13}{8}$ (D) $\frac{\sqrt{17}-16}{8}$

Q111. JEE Main 2022 (Online) 27th June Evening Shift

If m and n respectively are the number of local maximum and local minimum points of the function $f(x)=\int\limits_0^{x^2}\frac{t^2-5t+4}{2+e^t}dt$, then the ordered pair (m, n) is equal to

Open solution

(A) (3, 2) (B) (2, 3) (C) (2, 2) (D) (3, 4)

Q112. JEE Main 2022 (Online) 27th June Evening Shift

Let f be a differentiable function in $\left(0,\frac{\pi}{2}\right)$. If $\int\limits_{\cos x}^1 t^2 f(t)dt=\sin^3 x+\cos x$, then $\frac{1}{\sqrt{3}}f'\left(\frac{1}{\sqrt{3}}\right)$ is equal to

Open solution

(A) $6-9\sqrt{2}$ (B) $6-\frac{9}{\sqrt{2}}$ (C) $\frac{9}{2}-6\sqrt{2}$ (D) $\frac{9}{\sqrt{2}}-6$

Q113. JEE Main 2022 (Online) 27th June Evening Shift

The integral $\int\limits_0^1\frac{1}{7^{\left[\frac{1}{x}\right]}}dx$, where [.] denotes the greatest integer function, is equal to

Open solution

(A) $1+6\log_e\left(\frac{6}{7}\right)$ (B) $1-6\log_e\left(\frac{6}{7}\right)$
(C) $\log_e\left(\frac{7}{6}\right)$ (D) $1-7\log_e\left(\frac{6}{7}\right)$

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Q114. JEE Main 2022 (Online) 27th June Morning Shift

The value of the integral

$\int_{-2}^{2} \frac{|x^3+x|}{(e^{x|x|}+1)}dx$ is equal to :

(A) $5e^2$ (B) $3e^{-2}$ (C) 4 (D) 6

Open solution

Q115. JEE Main 2022 (Online) 25th June Evening Shift

If $b_n = \int_0^{\frac{\pi}{2}} \frac{\cos^2 nx}{\sin x}dx$, $n \in N$, then

(A) $b_3 - b_2$, $b_4 - b_3$, $b_5 - b_4$ are in A.P. with common difference -2

(B) $\frac{1}{b_3-b_2}, \frac{1}{b_4-b_3}, \frac{1}{b_5-b_4}$ are in an A.P. with common difference 2

(C) $b_3 - b_2$, $b_4 - b_3$, $b_5 - b_4$ are in a G.P.

(D) $\frac{1}{b_3-b_2}, \frac{1}{b_4-b_3}, \frac{1}{b_5-b_4}$ are in an A.P. with common difference -2

Open solution

Q116. JEE Main 2022 (Online) 25th June Morning Shift

The value of $\int_0^{\pi} \frac{e^{\cos x}\sin x}{(1+\cos^2 x)(e^{\cos x}+e^{-\cos x})}dx$ is equal to:

(A) $\frac{\pi^2}{4}$ (B) $\frac{\pi^2}{2}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Open solution

Q117. JEE Main 2022 (Online) 24th June Evening Shift

The value of the integral

$\int_{-\pi/2}^{\pi/2} \frac{dx}{(1+e^x)(\sin^6 x+\cos^6 x)}$ is equal to

(A) 2π (B) 0 (C) π (D) $\frac{\pi}{2}$

Open solution

Q118. JEE Main 2021 (Online) 1st September Evening Shift

Let f : R $\rightarrow$ R be a continuous function. Then $\lim\limits_{x\to\frac{\pi}{4}} \frac{\int_2^{\sec^2 x} f(x)\,dx}{x^2-\frac{\pi^2}{16}}$ is equal to :

(A) f (2) (B) 2f (2) (C) 2f $\left(\sqrt{2}\right)$ (D) 4f (2)

Open solution

Q119. JEE Main 2021 (Online) 1st September Evening Shift

Let $J_{n,m} = \int_0^{\frac{1}{2}} \frac{x^n}{x^m-1}dx$, $\forall$ n > m and n, m $\in$ N. Consider a matrix $A = [a_{ij}]_{3\times 3}$ where $a_{ij} = \begin{cases} J_{6+i,3} - J_{i+3,3}, & i \le j \\ 0, & i > j \end{cases}$. Then $\left|adjA^{-1}\right|$ is :

(A) $(15)^2 \times 2^{42}$ (B) $(15)^2 \times 2^{34}$ (C) $(105)^2 \times 2^{38}$ (D) $(105)^2 \times 2^{36}$

Open solution

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Q120. JEE Main 2021 (Online) 1st September Evening Shift

The function f(x), that satisfies the condition $f(x) = x + \int_0^{\pi/2} \sin x . \cos y \, f(y) \, dy$, is :

(A) $x + \frac{2}{3}(\pi - 2)\sin x$ (B) $x + (\pi + 2)\sin x$ (C) $x + \frac{\pi}{2}\sin x$ (D) $x + (\pi - 2)\sin x$

Open solution

Q121. JEE Main 2021 (Online) 31st August Evening Shift

If [x] is the greatest integer $\leq$ x, then

$\pi^2 \int_0^2 \left(\sin \frac{\pi x}{2}\right)(x - [x])^{[x]} dx$ is equal to :

(A) 2(π − 1) (B) 4(π − 1) (C) 4(π + 1) (D) 2(π + 1)

Open solution

Q122. JEE Main 2021 (Online) 31st August Morning Shift

Let f be a non-negative function in [0, 1] and twice differentiable in (0, 1). If $\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t)dt, 0 \leq x \leq 1$ and f(0) = 0, then $\lim_{x \to 0} \frac{1}{x^2} \int_0^x f(t)dt$:

(A) equals 0 (B) equals 1 (C) does not exist (D) equals $\frac{1}{2}$

Open solution

Q123. JEE Main 2021 (Online) 27th August Evening Shift

The value of the integral $\int_0^1 \frac{\sqrt{x}dx}{(1+x)(1+3x)(3+x)}$ is :

(A) $\frac{\pi}{8}\left(1 - \frac{\sqrt{3}}{2}\right)$ (B) $\frac{\pi}{4}\left(1 - \frac{\sqrt{3}}{6}\right)$

(C) $\frac{\pi}{8}\left(1 - \frac{\sqrt{3}}{6}\right)$ (D) $\frac{\pi}{4}\left(1 - \frac{\sqrt{3}}{2}\right)$

Open solution

Q124. JEE Main 2021 (Online) 27th August Morning Shift

$\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to :

(A) 6 (B) 8 (C) 5 (D) 10

Open solution

Q125. JEE Main 2021 (Online) 26th August Evening Shift

If the value of the integral $\int_0^5 \frac{x + [x]}{e^{x - [x]}} dx = \alpha e^{-1} + \beta$, where $\alpha, \beta \in$ R, 5α + 6β = 0, and [x] denotes the greatest integer less than or equal to x; then the value of $(\alpha + \beta)^2$ is equal to :

(A) 100 (B) 25 (C) 16 (D) 36

Open solution

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Q126. JEE Main 2021 (Online) 26th August Evening Shift

The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1+\sin^2 x}{1+\pi^{\sin x}}\right) dx$ is

(A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$

Open solution

Q127. JEE Main 2021 (Online) 26th August Morning Shift

The value of $\int_{\frac{-1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \left(\left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x-1}{x+1}\right)^2 - 2\right)^{\frac{1}{2}} dx$ is :

(A) $\log_e 4$ (B) $\log_e 16$

(C) $2\log_e 16$ (D) $4\log_e (3 + 2\sqrt{2})$

Open solution

Q128. JEE Main 2021 (Online) 27th July Evening Shift

Let f : (a, b) $\to$ R be twice differentiable function such that $f(x) = \int_a^x g(t)dt$ for a differentiable function g(x). If f(x) = 0 has exactly five distinct roots in (a, b), then g(x)g'(x) = 0 has at least :

(A) twelve roots in (a, b) (B) five roots in (a, b)

(C) seven roots in (a, b) (D) three roots in (a, b)

Open solution

Q129. JEE Main 2021 (Online) 27th July Morning Shift

The value of the definite integral

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{dx}{(1+e^{x\cos x})(\sin^4 x+\cos^4 x)}$ is equal to :

(A) $-\frac{\pi}{2}$ (B) $\frac{\pi}{2\sqrt{2}}$ (C) $-\frac{\pi}{4}$ (D) $\frac{\pi}{\sqrt{2}}$

Open solution

Q130. JEE Main 2021 (Online) 25th July Evening Shift

If $f(x) = \begin{cases} \int_0^x (5 + |1 - t|)dt, & x > 2 \\ 5x + 1, & x \le 2 \end{cases}$, then

(A) f(x) is not continuous at x = 2

(B) f(x) is everywhere differentiable

(C) f(x) is continuous but not differentiable at x = 2

(D) f(x) is not differentiable at x = 1

Open solution

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Q131. JEE Main 2021 (Online) 25th July Evening Shift

The value of the

integral $\int_{-1}^{1} \log\left(x+\sqrt{x^2+1}\right)dx$ is :

(A) 2 (B) 0 (C) -1 (D) 1

Open solution

Q132. JEE Main 2021 (Online) 25th July Morning Shift

The value of the definite integral $\int_{\pi/24}^{5\pi/24} \frac{dx}{1+\sqrt[3]{\tan 2x}}$ is :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{18}$

Open solution

Q133. JEE Main 2021 (Online) 25th July Morning Shift

Let $f:[0,\infty)\to[0,\infty)$ be defined as $f(x)=\int_0^x [y]dy$

where [x] is the greatest integer less than or equal to x. Which of the following is true?

(A) f is continuous at every point in $[0,\infty)$ and differentiable except at the integer points.

(B) f is both continuous and differentiable except at the integer points in $[0,\infty)$.

(C) f is continuous everywhere except at the integer points in $[0,\infty)$.

(D) f is differentiable at every point in $[0,\infty)$.

Open solution

Q134. JEE Main 2021 (Online) 22th July Evening Shift

If $\int_0^{100\pi} \frac{\sin^2 x}{e^{\left(\frac{x}{\pi}-\left[\frac{x}{\pi}\right]\right)}}dx = \frac{\alpha\pi^3}{1+4\pi^2}, \alpha\in R$ where [x] is the greatest integer less than or equal to x, then the value of α is :

(A) 200 $(1-e^{-1})$ (B) 100 $(1-e)$ (C) 50 $(e-1)$ (D) 150 $(e^{-1}-1)$

Open solution

Q135. JEE Main 2021 (Online) 20th July Evening Shift

If [x] denotes the greatest integer less than or equal to x, then the value of the integral $\int_{-\pi/2}^{\pi/2}[[x]-\sin x]dx$ is equal to :

(A) $-\pi$ (B) π (C) 0 (D) 1

Open solution

Q136. JEE Main 2021 (Online) 20th July Evening Shift

If the real part of the complex number $(1-\cos\theta+2i\sin\theta)^{-1}$ is $\frac{1}{5}$ for $\theta\in(0,\pi)$, then the value of the integral $\int_0^{\theta}\sin x dx$ is equal to:

(A) 1 (B) 2 (C) -1 (D) 0

Open solution

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Q137. JEE Main 2021 (Online) 20th July Evening Shift

If $f : R \to R$ is given by $f(x) = x + 1$, then the value of $\lim_{n\to\infty} \frac{1}{n}\left[f(0) + f\left(\frac{5}{n}\right) + f\left(\frac{10}{n}\right) + \ldots\ldots + f\left(\frac{5(n-1)}{n}\right)\right]$ is :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{1}{2}$ (D) $\frac{7}{2}$

Open solution

Q138. JEE Main 2021 (Online) 20th July Evening Shift

Let $g(t) = \int_{-\pi/2}^{\pi/2} \cos\left(\frac{\pi}{4}t + f(x)\right)dx$, where $f(x) = \log_e\left(x + \sqrt{x^2+1}\right), x \in R$. Then which one of the following is correct?

(A) g(1) = g(0) (B) $\sqrt{2}g(1) = g(0)$ (C) $g(1) = \sqrt{2}g(0)$ (D) g(1) + g(0) = 0

Open solution

Q139. JEE Main 2021 (Online) 20th July Morning Shift

Let a be a positive real number such that $\int_0^a e^{x-[x]}dx = 10e - 9$ where [x] is the greatest integer less than or equal to x. Then a is equal to:

(A) $10 - \log_e(1+e)$ (B) $10 + \log_e 2$ (C) $10 + \log_e 3$ (D) $10 + \log_e(1+e)$

Open solution

Q140. JEE Main 2021 (Online) 20th July Morning Shift

The value of the integral $\int_{-1}^{1} \log_e(\sqrt{1-x} + \sqrt{1+x})dx$ is equal to:

(A) $\frac{1}{2}\log_e 2 + \frac{\pi}{4} - \frac{3}{2}$ (B) $2\log_e 2 + \frac{\pi}{4} - 1$

(C) $\log_e 2 + \frac{\pi}{2} - 1$ (D) $2\log_e 2 + \frac{\pi}{2} - \frac{1}{2}$

Open solution

Q141. JEE Main 2021 (Online) 18th March Evening Shift

Let g(x) = $\int_0^x f(t)dt$, where f is continuous function in [0, 3] such that $\frac{1}{3} \le$ f(t) ≤ 1 for all t$\in$ [0, 1] and 0 $\le$ f(t) $\le \frac{1}{2}$ for all t$\in$ (1, 3]. The largest possible interval in which g(3) lies is :

(A) $\left[-1, -\frac{1}{2}\right]$ (B) $\left[-\frac{3}{2}, -1\right]$

(C) [1, 3] (D) $\left[\frac{1}{3}, 2\right]$

Open solution

Q142. JEE Main 2021 (Online) 17th March Evening Shift

Let f : R $\to$ R be defined as f(x) = e^{-x}sinx. If F : [0, 1] $\to$ R is a differentiable function with that F(x) = $\int_0^x f(t)dt$, then the value of $\int_0^1 (F'(x) + f(x))e^x dx$ lies in the interval

(A) $\left[\frac{331}{360}, \frac{334}{360}\right]$ (B) $\left[\frac{330}{360}, \frac{331}{360}\right]$

(C) $\left[\frac{335}{360}, \frac{336}{360}\right]$ (D) $\left[\frac{327}{360}, \frac{329}{360}\right]$

Open solution

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Q143. JEE Main 2021 (Online) 17th March Evening Shift

If the integral

$\int_0^{10} \frac{[\sin 2\pi x]}{e^{x-[x]}}dx = \alpha e^{-1} + \beta e^{-\frac{1}{2}} + \gamma$, where α, β, γ are integers and [x] denotes the greatest integer less than or equal to x, then the value of $\alpha + \beta + \gamma$ is equal to :

(A) 0 (B) 10 (C) 20 (D) 25

Open solution

Q144. JEE Main 2021 (Online) 17th March Morning Shift

Which of the following statements is correct for the function g(α) for $\alpha \in$ R such that

$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^\alpha x}{\cos^\alpha x + \sin^\alpha x}dx$

(A) $g(\alpha)$ is a strictly increasing function

(B) $g(\alpha)$ is an even function

(C) $g(\alpha)$ has an inflection point at $\alpha = -\frac{1}{2}$

(D) $g(\alpha)$ is a strictly decreasing function

Open solution

Q145. JEE Main 2021 (Online) 16th March Evening Shift

Consider the integral
$I = \int_0^{10} \frac{[x]e^{[x]}}{e^{x-1}}dx$,
where [x] denotes the greatest integer less than or equal to x. Then the value of I is equal to :

(A) 45 (e $-$ 1) (B) 45 (e + 1) (C) 9 (e + 1) (D) 9 (e $-$ 1)

Open solution

Q146. JEE Main 2021 (Online) 16th March Evening Shift

Let P(x) = x^2 + bx + c be a quadratic polynomial with real coefficients such that $\int_0^1 P(x)dx$ = 1 and P(x) leaves remainder 5 when it is divided by (x $-$ 2). Then the value of 9(b + c) is equal to :

(A) 9 (B) 11 (C) 7 (D) 15

Open solution

Q147. JEE Main 2021 (Online) 26th February Evening Shift

Let $f(x) = \int_0^x e^t f(t)dt + e^x$ be a differentiable function for all x$\in$R. Then f(x) equals :

(A) $e^{(e^{x-1})}$ (B) $2e^{e^x} - 1$ (C) $2e^{e^x-1} - 1$ (D) $e^{e^x} - 1$

Open solution

Q148. JEE Main 2021 (Online) 26th February Evening Shift

For x > 0, if $f(x) = \int_1^x \frac{\log_e t}{(1+t)}dt$, then $f(e) + f\left(\frac{1}{e}\right)$ is equal to :

(A) $\frac{1}{2}$ (B) -1 (C) 0 (D) 1

Open solution

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Q149. JEE Main 2021 (Online) 26th February Morning Shift

The value of $\int_{-\pi/2}^{\pi/2} \frac{\cos^2 x}{1+3^x}\,dx$ is :

(A) 2π (B) $\frac{\pi}{2}$ (C) 4π (D) $\frac{\pi}{4}$

Open solution

Q150. JEE Main 2021 (Online) 26th February Morning Shift

The value of $\sum_{n=1}^{100} \int_{n-1}^{n} e^{x-[x]}dx$, where [x] is the greatest integer $\leq$ x, is :

(A) 100e (B) 100(e $-$ 1) (C) 100(1 + e) (D) 100(1 $-$ e)

Open solution

Q151. JEE Main 2021 (Online) 25th February Evening Shift

If $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x\,dx$, then :

(A) $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$ are in A.P.

(B) $I_2 + I_4, I_3 + I_5, I_4 + I_6$ are in A.P.

(C) $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$ are in G.P.

(D) $I_2 + I_4, (I_3 + I_5)^2, I_4 + I_6$ are in G.P.

Open solution

Q152. JEE Main 2021 (Online) 25th February Morning Shift

The value of $\int_{-1}^{1} x^2 e^{[x^3]}dx$, where [t] denotes the greatest integer $\leq$ t, is :

(A) $\frac{e+1}{3}$ (B) $\frac{e-1}{3e}$ (C) $\frac{1}{3e}$ (D) $\frac{e+1}{3e}$

Open solution

Q153. JEE Main 2021 (Online) 24th February Evening Shift

The value of the integral, $\int_{1}^{3} [x^2 - 2x - 2]dx$, where [x] denotes the greatest integer less than or equal to x, is :

(A) -5 (B) $-\sqrt{2} - \sqrt{3} + 1$ (C) -4 (D) $-\sqrt{2} - \sqrt{3} - 1$

Open solution

Q154. JEE Main 2021 (Online) 24th February Evening Shift

Let f(x) be a differentiable function defined on [0, 2] such that f'(x) = f'(2 $-$ x) for all x$\in$ (0, 2), f(0) = 1 and f(2) = e^2. Then the value of $\int_{0}^{2} f(x)dx$ is :

(A) 1 + e^2 (B) 2(1 + e^2) (C) 1 $-$ e^2 (D) 2(1 $-$ e^2)

Open solution

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Q155. JEE Main 2021 (Online) 24th February Evening Shift

Let f be a twice differentiable function defined on R such that f(0) = 1, f'(0) = 2 and f'(x) $\neq$ 0 for all x $\in$ R. If $\begin{vmatrix} f(x) & f'(x) \\ f'(x) & f''(x) \end{vmatrix}$ = 0, for all x$\in$R, then the value of f(1) lies in the interval :

(A) (0, 3) (B) (9, 12) (C) (3, 6) (D) (6, 9)

Open solution

Q156. JEE Main 2021 (Online) 24th February Morning Shift

$\lim\limits_{x\to 0} \frac{\int_0^{x^2} \left(\sin\sqrt{t}\right)dt}{x^3}$ is equal to :

(A) $\frac{1}{15}$ (B) 0 (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

Open solution

Q157. JEE Main 2020 (Online) 6th September Evening Slot

The integral $\int_1^2 e^x . x^x \left(2 + \log_e x\right)dx$ equals :

(A) e(4e + 1) (B) e(2e − 1) (C) e(4e − 1) (D) $4e^2 - 1$

Open solution

Q158. JEE Main 2020 (Online) 6th September Morning Slot

$\lim\limits_{x\to 1} \left(\frac{\int_0^{(x-1)^2} t\cos\left(t^2\right)dt}{(x-1)\sin(x-1)} \right)$

(A) is equal to 0 (B) is equal to $\frac{1}{2}$

(C) does not exist (D) is equal to $-\frac{1}{2}$

Open solution

Q159. JEE Main 2020 (Online) 6th September Morning Slot

If $I_1 = \int_0^1 \left(1 - x^{50}\right)^{100} dx$ and

$I_2 = \int_0^1 \left(1 - x^{50}\right)^{101} dx$ such

that $I_2 = \alpha I_1$ then α equals to :

(A) $\frac{5051}{5050}$ (B) $\frac{5050}{5051}$ (C) $\frac{5050}{5049}$ (D) $\frac{5049}{5050}$

Open solution

Q160. JEE Main 2020 (Online) 5th September Morning Slot

The value of $\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+e^{\sin x}} dx$ is:

(A) π (B) $\frac{3\pi}{2}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Open solution

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Q161. JEE Main 2020 (Online) 4th September Evening Slot

The integral

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \tan^3 x . \sin^2 3x \left(2\sec^2 x . \sin^2 3x + 3\tan x . \sin 6x\right) dx$$

is equal to:

Open solution

(A) $-\frac{1}{9}$ (B) $-\frac{1}{18}$ (C) $\frac{7}{18}$ (D) $\frac{9}{2}$

Q162. JEE Main 2020 (Online) 4th September Morning Slot

Let $f(x) = |x - 2|$ and g(x) = f(f(x)), $x \in [0, 4]$. Then

$\int_0^3 (g(x) - f(x))dx$ is equal to:

Open solution

(A) 1 (B) 0 (C) $\frac{1}{2}$ (D) $\frac{3}{2}$

Q163. JEE Main 2020 (Online) 3rd September Evening Slot

If the value of the integral

$$\int_0^{\frac{1}{2}} \frac{x^2}{(1-x^2)^{\frac{3}{2}}} dx$$

is $\frac{k}{6}$, then k is equal to :

Open solution

(A) $2\sqrt{3} + \pi$ (B) $3\sqrt{2} - \pi$ (C) $3\sqrt{2} + \pi$ (D) $2\sqrt{3} - \pi$

Q164. JEE Main 2020 (Online) 3rd September Morning Slot

$\int_{-\pi}^{\pi} |\pi - |x||dx$ is equal to :

Open solution

(A) π^2 (B) $2\pi^2$ (C) $\sqrt{2}\pi^2$ (D) $\frac{\pi^2}{2}$

Q165. JEE Main 2020 (Online) 9th January Evening Slot

Let a function f : [0, 5] $\rightarrow$ R be continuous, f(1) = 3 and F be defined as :

Open solution

$F(x) = \int_1^x t^2 g(t)dt$, where $g(t) = \int_1^t f(u)du$

Then for the function F, the point x = 1 is :

(A) a point of inflection. (B) a point of local maxima.

(C) a point of local minima. (D) not a critical point.

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Q166. JEE Main 2020 (Online) 9th January Morning Slot

The value of $\int_0^{2\pi} \frac{x\sin^8 x}{\sin^8 x+\cos^8 x}dx$ is equal to :

(A) 4π (B) 2π (C) π^2 (D) $2\pi^2$

Open solution

Q167. JEE Main 2020 (Online) 9th January Morning Slot

If for all real triplets (a, b, c), f(x) = a + bx + cx^2; then $\int_0^1 f(x)dx$ is equal to :

(A) $\frac{1}{6}\left\{f(0)+f(1)+4f\left(\frac{1}{2}\right)\right\}$ (B) $2\left\{3f(1)+2f\left(\frac{1}{2}\right)\right\}$

(C) $\frac{1}{3}\left\{f(0)+f\left(\frac{1}{2}\right)\right\}$ (D) $\frac{1}{2}\left\{f(1)+3f\left(\frac{1}{2}\right)\right\}$

Open solution

Q168. JEE Main 2020 (Online) 8th January Evening Slot

If $I=\int_1^2 \frac{dx}{\sqrt{2x^3-9x^2+12x+4}}$, then :

(A) $\frac{1}{16} < I^2 < \frac{1}{9}$ (B) $\frac{1}{8} < I^2 < \frac{1}{4}$

(C) $\frac{1}{9} < I^2 < \frac{1}{8}$ (D) $\frac{1}{6} < I^2 < \frac{1}{2}$

Open solution

Q169. JEE Main 2020 (Online) 8th January Evening Slot

$\lim_{x\to 0} \frac{\int_0^x t\sin(10t)dt}{x}$ is equal to

(A) $-\frac{1}{5}$ (B) $-\frac{1}{10}$ (C) 0 (D) $\frac{1}{10}$

Open solution

Q170. JEE Main 2020 (Online) 7th January Evening Slot

The value of α for which

$4\alpha\int_{-1}^{2} e^{-\alpha|x|}dx = 5$, is:

(A) $\log_e 2$ (B) $\log_e\sqrt{2}$

(C) $\log_e\left(\frac{4}{3}\right)$ (D) $\log_e\left(\frac{3}{2}\right)$

Open solution

Q171. JEE Main 2020 (Online) 7th January Evening Slot

If θ_1 and θ_2 be respectively the smallest and the largest values of θ in (0, 2π) - {π} which satisfy the equation,

2cot$^2\theta$ - $\frac{5}{\sin\theta}$ + 4 = 0, then

$\int_{\theta_1}^{\theta_2} \cos^2 3\theta d\theta$ is equal to :

(A) $\frac{\pi}{9}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{3}+\frac{1}{6}$

Open solution

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Q172. JEE Main 2020 (Online) 7th January Morning Slot

If ƒ(a + b + 1 - x) = ƒ(x), for all x, where a and b are fixed positive real numbers, then

$\frac{1}{a+b}\int_a^b x\,(f(x)+f(x+1))dx$ is equal to:

(A) $\int_{a-1}^{b-1} f(x+1)dx$

(B) $\int_{a+1}^{b+1} f(x+1)dx$

(C) $\int_{a-1}^{b-1} f(x)dx$

(D) $\int_{a+1}^{b+1} f(x)dx$

Open solution

Q173. JEE Main 2019 (Online) 12th April Evening Slot

A value of α such that

$\int_{\alpha}^{\alpha+1} \frac{dx}{(x+\alpha)(x+\alpha+1)} = \log_e\left(\frac{9}{8}\right)$ is :

(A) 2

(B) - 2

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

Open solution

Q174. JEE Main 2019 (Online) 12th April Morning Slot

If $\int_0^{\frac{\pi}{2}} \frac{\cot x}{\cot x+\operatorname{cosec} x}dx$ = m(π + n), then m.n is equal to

(A) - 1

(B) 1

(C) $-\frac{1}{2}$

(D) $\frac{1}{2}$

Open solution

Q175. JEE Main 2019 (Online) 12th April Morning Slot

Let f : R $\rightarrow$ R be a continuously differentiable function such that f(2) = 6 and f'(2) = $\frac{1}{48}$. If $\int_6^{f(x)} 4t^3dt$ = (x - 2)g(x), then $\lim_{x\to 2} g(x)$ is equal to :

(A) 18

(B) 36

(C) 12

(D) 24

Open solution

Q176. JEE Main 2019 (Online) 10th April Evening Slot

The integral $\int_{\pi/6}^{\pi/3} \sec^{2/3}x\,\operatorname{cosec}^{4/3}x dx$ is equal to :

(A) $3^{\frac{5}{3}}-3^{\frac{1}{3}}$

(B) $3^{\frac{5}{6}}-3^{\frac{2}{3}}$

(C) $3^{\frac{4}{3}}-3^{\frac{1}{3}}$

(D) $3^{\frac{7}{6}}-3^{\frac{5}{6}}$

Open solution

Q177. JEE Main 2019 (Online) 10th April Morning Slot

The value of $\int_0^{2\pi} [\sin 2x\,(1+\cos 3x)]dx$,

where [t] denotes the greatest integer function is :

(A) 2π

(B) π

(C) -2π

(D) $-\pi$

Open solution

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Q178. JEE Main 2019 (Online) 9th April Evening Slot

If f : R $\rightarrow$ R is a differentiable function and f(2) = 6,

then $\lim_{x\to 2} \frac{\int_{6}^{f(x)} 2tdt}{(x-2)}$ is :-

Open solution

(A) 2f'(2) (B) 24f'(2) (C) 0 (D) 12f'(2)

Q179. JEE Main 2019 (Online) 9th April Evening Slot

The value of the integral $\int_0^1 x\cot^{-1}(1-x^2+x^4)dx$ is :-

Open solution

(A) $\frac{\pi}{2}-\frac{1}{2}\log_e 2$ (B) $\frac{\pi}{4}-\log_e 2$

(C) $\frac{\pi}{4}-\frac{1}{2}\log_e 2$ (D) $\frac{\pi}{2}-\log_e 2$

Q180. JEE Main 2019 (Online) 9th April Morning Slot

The value of $\int_0^{\pi/2} \frac{\sin^3 x}{\sin x+\cos x}dx$ is

Open solution

(A) $\frac{\pi-2}{8}$ (B) $\frac{\pi-2}{4}$ (C) $\frac{\pi-1}{2}$ (D) $\frac{\pi-1}{4}$

Q181. JEE Main 2019 (Online) 8th April Evening Slot

Let $f(x)=\int_0^x g(t)dt$ where g is a non-zero even function. If f(x + 5) = g(x), then $\int_0^x f(t)dt$ equals-

Open solution

(A) $5\int_{x+5}^{5} g(t)dt$ (B) $\int_{x+5}^{5} g(t)dt$

(C) $\int_{5}^{x+5} g(t)dt$ (D) $2\int_{5}^{x+5} g(t)dt$

Q182. JEE Main 2019 (Online) 8th April Morning Slot

If $f(x)=\frac{2-x\cos x}{2+x\cos x}$ and g(x) = $\log_e$x, (x > 0) then the value of integral

Open solution

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} g\left(f\left(x\right)\right)dx$ is

(A) $\log_e 3$ (B) $\log_e 2$ (C) $\log_e 1$ (D) $\log_e e$

Q183. JEE Main 2019 (Online) 12th January Evening Slot

The integral $\int_1^e \left\{\left(\frac{x}{e}\right)^{2x}-\left(\frac{e}{x}\right)^{x}\right\}$ $\log_e$ x dx is equal to :

Open solution

(A) $-\frac{1}{2}+\frac{1}{e}-\frac{1}{2e^2}$ (B) $\frac{3}{2}-e-\frac{1}{2e^2}$

(C) $\frac{1}{2}-e-\frac{1}{e^2}$ (D) $\frac{3}{2}-\frac{1}{e}-\frac{1}{2x^2}$

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Q184. JEE Main 2019 (Online) 12th January Morning Slot

Let f and g be continuous functions on [0, a] such that f(x) = f(a − x) and g(x) + g(a − x) = 4, then $\int_0^a f(x)\,g(x)\,dx$ is equal to :

Open solution

(A) $4\int_0^a f(x)dx$

(B) $-3\int_0^a f(x)dx$

(C) $\int_0^a f(x)dx$

(D) $2\int_0^a f(x)dx$

Q185. JEE Main 2019 (Online) 11th January Evening Slot

The integral $\int_{\pi/6}^{\pi/4} \frac{dx}{\sin 2x\left(\tan^5 x+\cot^5 x\right)}$ equals :

Open solution

(A) $\frac{\pi}{40}$

(B) $\frac{1}{20}\tan^{-1}\left(\frac{1}{9\sqrt{3}}\right)$

(C) $\frac{1}{10}\left(\frac{\pi}{4}-\tan^{-1}\left(\frac{1}{9\sqrt{3}}\right)\right)$

(D) $\frac{1}{5}\left(\frac{\pi}{4}-\tan^{-1}\left(\frac{1}{3\sqrt{3}}\right)\right)$

Q186. JEE Main 2019 (Online) 11th January Morning Slot

The value of the integral $\int_{-2}^{2} \frac{\sin^2 x}{\left[\frac{x}{\pi}\right]+\frac{1}{2}}\,dx$ (where [x] denotes the greatest integer less than or equal to x) is

Open solution

(A) 0 (B) 4 (C) 4− sin 4 (D) sin 4

Q187. JEE Main 2019 (Online) 10th January Evening Slot

The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{[x]+[\sin x]+4}$, where [t] denotes the greatest integer less than or equal to t, is

Open solution

(A) $\frac{1}{12}(7\pi-5)$

(B) $\frac{1}{12}(7\pi+5)$

(C) $\frac{3}{10}(4\pi-3)$

(D) $\frac{3}{20}(4\pi-3)$

Q188. JEE Main 2019 (Online) 10th January Evening Slot

If $\int_0^x f(t)\,dt = x^2 + \int_x^1 t^2 f(t)\,dt$ then $f'\left(\frac{1}{2}\right)$ is -

Open solution

(A) $\frac{18}{25}$ (B) $\frac{6}{25}$ (C) $\frac{24}{25}$ (D) $\frac{4}{5}$

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Q189. JEE Main 2019 (Online) 10th January Morning Slot

Let $I = \int_a^b (x^4 - 2x^2)dx$. If I is minimum then the ordered pair (a, b) is -

(A) $(\sqrt{2}, -\sqrt{2})$ (B) $(0, \sqrt{2})$

(C) $(-\sqrt{2}, \sqrt{2})$ (D) $(-\sqrt{2}, 0)$

Open solution

Q190. JEE Main 2019 (Online) 9th January Evening Slot

Let f be a differentiable function from

R to **R** such that $|f(x) - f(y)| \leq 2|x - y|^{\frac{3}{2}}$,

for all $x, y \in$ **R**.

If $f(0) = 1$

then $\int_0^1 f^2(x)dx$ is equal to :

(A) 1 (B) 2 (C) $\frac{1}{2}$ (D) 0

Open solution

Q191. JEE Main 2019 (Online) 9th January Evening Slot

If $\int_0^{\frac{\pi}{3}} \frac{\tan\theta}{\sqrt{2k\sec\theta}} d\theta = 1 - \frac{1}{\sqrt{2}}, (k > 0)$, then value of k is :

(A) 4 (B) $\frac{1}{2}$ (C) 1 (D) 2

Open solution

Q192. JEE Main 2019 (Online) 9th January Morning Slot

The value of $\int_0^{\pi} |\cos x|^3 dx$ is :

(A) $\frac{4}{3}$ (B) $-\frac{4}{3}$ (C) 0 (D) $\frac{2}{3}$

Open solution

Q193. JEE Main 2018 (Online) 16th April Morning Slot

If $f(x) = \int_0^x t(\sin x - \sin t)dt$ then :

(A) f'''(x) + f''(x) = sinx (B) f'''(x) + f''(x) − f'(x) = cosx

(C) f'''(x) + f'(x) = cosx − 2x sinx (D) f'''(x) − f''(x) = cosx − 2x sinx

Open solution

Q194. JEE Main 2018 (Offline)

The value of $\int_{-\pi/2}^{\pi/2} \frac{\sin^2 x}{1+2^x} dx$ is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{8}$ (C) $\frac{\pi}{2}$ (D) 4π

Open solution

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Q195. JEE Main 2018 (Online) 15th April Evening Slot

The value of integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{x}{1+\sin x}dx$ is :

(A) $\pi\sqrt{2}$ (B) $\pi\left(\sqrt{2}-1\right)$

(C) $\frac{\pi}{2}\left(\sqrt{2}+1\right)$ (D) $2\pi\left(\sqrt{2}-1\right)$

Open solution

Q196. JEE Main 2018 (Online) 15th April Evening Slot

If $I_1 = \int_0^1 e^{-x}\cos^2 x\,dx$;

$I_2 = \int_0^1 e^{-x^2}\cos^2 x\,dx$ and

$I_3 = \int_0^1 e^{-x^3}dx$; then

(A) $I_2 > I_3 > I_1$ (B) $I_2 > I_1 > I_3$ (C) $I_3 > I_2 > I_1$ (D) $I_3 > I_1 > I_2$

Open solution

Q197. JEE Main 2018 (Online) 15th April Morning Slot

The value of the integral

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^4 x\left(1+\log\left(\frac{2+\sin x}{2-\sin x}\right)\right)dx$ is :

(A) 0 (B) $\frac{3}{4}$ (C) $\frac{3}{8}\pi$ (D) $\frac{3}{16}\pi$

Open solution

Q198. JEE Main 2017 (Online) 9th April Morning Slot

If $\int_1^2 \frac{dx}{(x^2-2x+4)^{\frac{3}{2}}} = \frac{k}{k+5}$, then k is equal to :

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q199. JEE Main 2017 (Online) 8th April Morning Slot

The integral $\int_{\frac{\pi}{12}}^{\frac{\pi}{4}} \frac{8\cos 2x}{(\tan x+\cot x)^3}dx$ equals :

(A) $\frac{15}{128}$ (B) $\frac{15}{64}$ (C) $\frac{13}{32}$ (D) $\frac{13}{256}$

Open solution

Q200. JEE Main 2017 (Offline)

The integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dx}{1+\cos x}$ is equal to

(A) 2 (B) 4 (C) -1 (D) -2

Open solution

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Q201. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

The value of the integral

$$\int_4^{10} \frac{[x^2]\,dx}{[x^2-28x+196]+[x^2]},$$

where [x] denotes the greatest integer less than or equal to x, is :

(A) 6 (B) 3 (C) 7 (D) $\frac{1}{3}$

Q202. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

For $x \in \mathbf{R}$, $x \neq 0$, if y(x) is a differentiable function such that

$x\int_1^x y(t)\,dt = (x+1)\int_1^x ty(t)\,dt$, then y (x) equals :

(where C is a constant.)

(A) $\frac{C}{x}e^{-\frac{1}{x}}$ (B) $\frac{C}{x^2}e^{-\frac{1}{x}}$

(C) $\frac{C}{x^3}e^{-\frac{1}{x}}$ (D) $Cx^3\,\frac{1}{e^x}$

Q203. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

If $2\int_0^1 \tan^{-1}x\,dx = \int_0^1 \cot^{-1}\left(1-x+x^2\right)dx,$

then $\int_0^1 \tan^{-1}\left(1-x+x^2\right)dx$ is equalto :

(A) log4 (B) $\frac{\pi}{2}$ + log2 (C) log2 (D) $\frac{\pi}{2}$ − log4

Q204. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

Let f : **R** → **R** be a function such that f(2 − x) = f (2 + x) and

f(4 − x) = f (4 + x), for all $x \in \mathbf{R}$ and $\int_0^2 f(x)dx = 5$.

Then the value of $\int_{10}^{50} f(x)\,dx$ is :

(A) 80 (B) 100 (C) 125 (D) 200

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Q205. JEE Main 2015 (Online) 11th April Morning Slot

Let f : $(-1, 1) \to \mathbf{R}$ be a continuous function.

If $\int_0^{\sin x} f(t)\,dt = \frac{\sqrt{3}}{2}x$,

then $f\left(\frac{\sqrt{3}}{2}\right)$ is equal to :

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{3}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{1}{2}$

Open solution

Q206. JEE Main 2015 (Online) 10th April Morning Slot

For x > 0, Let f(x) = $\int_1^x \frac{\log t}{1+t}\,dt$. Then f(x) + f$\left(\frac{1}{x}\right)$ is equal to :

(A) $\frac{1}{4}(\log x)^2$ (B) $\frac{1}{2}(\log x)^2$

(C) $\log x$ (D) $\frac{1}{4}\log x^2$

Open solution

Q207. JEE Main 2015 (Offline)

The integral

$\int_2^4 \frac{\log x^2}{\log x^2 + \log(36 - 12x + x^2)}\,dx$ is equal to :

(A) 1 (B) 6 (C) 2 (D) 4

Open solution

Q208. JEE Main 2014 (Online) 19th April Morning Slot

Let function F be defined as $F(x) = \int_1^x \frac{e^t}{t}dt,\ x > 0$

then the value of the integral $\int_1^x \frac{e^t}{t+a}dt$, where a > 0, is :

(A) e^a[F(x) − F(1+a)] (B) e^{-a} [F(x + a) − F(a)]

(C) e^a[F(x + a) − F(1 + a)] (D) e^{-a}[F(x + a) − F(1 + a)]

Open solution

Q209. JEE Main 2014 (Online) 12th April Morning Slot

If [ ] denotes the greatest integer function then the integral $\int_0^{\pi} [\cos x]\,dx$ is equal to :

(A) $\frac{\pi}{2}$ (B) 0 (C) − 1 (D) $-\frac{\pi}{2}$

Open solution

Q210. JEE Main 2014 (Online) 12th April Morning Slot

If for a continuous function f(x), $\int_{-\pi}^{t} (f(x) + x)dx = \pi^2 - t^2$,

for all $t \geq -\pi$, then $f\left(-\frac{\pi}{3}\right)$ is equal to :

(A) π (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Open solution

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Q211. JEE Main 2014 (Online) 11th April Morning Slot

If for $n \geq 1$, $P_n = \int_1^e (\log x)^n dx$,

then $P_{10} - 90P_8$ is equal to :

(A) -9 (B) 10e (C) $-9e$ (D) 10

Open solution

Q212. JEE Main 2014 (Online) 9th April Morning Slot

The integral $\int_0^{\frac{1}{2}} \frac{\ln(1+2x)}{1+4x^2} dx$, equals :

(A) $\frac{\pi}{4} \ln 2$ (B) $\frac{\pi}{8} \ln 2$ (C) $\frac{\pi}{16} \ln 2$ (D) $\frac{\pi}{32} \ln 2$

Open solution

Q213. JEE Main 2014 (Offline)

The integral $\int_0^{\pi} \sqrt{1 + 4\sin^2\frac{x}{2} - 4\sin\frac{x}{2}} dx$ equals:

(A) $4\sqrt{3} - 4$ (B) $4\sqrt{3} - 4 - \frac{\pi}{3}$
(C) $\pi - 4$ (D) $\frac{2\pi}{3} - 4 - 4\sqrt{3}$

Open solution

Q214. JEE Main 2013 (Online) 25th April Morning Slot

For $0 \leq x \leq \frac{\pi}{2}$, the value of

$\int_0^{\sin^2 x} \sin^{-1}\left(\sqrt{t}\right) dt + \int_0^{\cos^2 x} \cos^{-1}\left(\sqrt{t}\right) dt$ equals :

(A) $\frac{\pi}{4}$ (B) 0 (C) 1 (D) $-\frac{\pi}{4}$

Open solution

Q215. JEE Main 2013 (Online) 25th April Morning Slot

Let $f : [-2, 3] \to [0, \infty)$ be a continuous function such that $f(1 - x) = f(x)$ for all $x \in [-2, 3]$. If R_1 is the numerical value of the area of the region bounded by $y = f(x)$, $x = -2$, $x = 3$ and the axis of x and $R_2 = \int_{-2}^{3} xf(x)dx$, then :

(A) $3R_1 = 2R_2$ (B) $2R_1 = 3R_2$ (C) $R_1 = R_2$ (D) $R_1 = 2R_2$

Open solution

Q216. JEE Main 2013 (Online) 23th April Morning Slot

The value of $\int_{-\pi/2}^{\pi/2} \frac{\sin^2 x}{1+2^x} dx$ is :

(A) π (B) $\frac{\pi}{2}$ (C) 4π (D) $\frac{\pi}{4}$

Open solution

Q217. JEE Main 2013 (Online) 22th April Morning Slot

The integral $\int_{7\pi/4}^{7\pi/3} \sqrt{\tan^2 x} dx$ is equal to :

(A) log $2\sqrt{2}$ (B) log 2 (C) 2 log 2 (D) log$\sqrt{2}$

Open solution

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Q218. JEE Main 2013 (Online) 9th April Morning Slot

If $x = \int_0^y \frac{dt}{\sqrt{1+t^2}}$, then $\frac{d^2y}{dx^2}$ is equal to :

(A) y (B) $\sqrt{1+y^2}$ (C) $\frac{x}{\sqrt{1+y^2}}$ (D) y^2

Open solution

Q219. JEE Main 2013 (Offline)

Statement-1 : The value of the integral $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$ is equal to $\pi/6$

Statement-2 : $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$.

(A) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

(C) Statement- 1 is true; Statement-2 is False.

(D) Statement-1 is false; Statement-2 is true.

Open solution

Q220. AIEEE 2012 (Online) 26th May

If $[x]$ is the greatest integer $\leq x$, then the value of the integral $\int_{-0.9}^{0.9} \left([x^2] + \log\left(\frac{2-x}{2+x}\right)\right)dx$ is

(A) 0.486 (B) 0.243 (C) 1.8 (D) 0

Open solution

Q221. AIEEE 2012 (Online) 19th May

The value of the integral $\int_0^{0.9} [x - 2[x]]\, dx$, where $[.]$ denotes the greatest integer function is

(A) 0.9 (B) 1.8 (C) -0.9 (D) 0

Open solution

Q222. AIEEE 2012 (Online) 12th May

If $\frac{d}{dx}G(x) = \frac{e^{\tan x}}{x}$, $x \in (0, \pi/2)$, then $\int_{1/4}^{1/2} \frac{2}{x} \cdot e^{\tan(\pi x^2)}dx$ is equal to

(A) $G(\pi/4) - G(\pi/16)$ (B) $2[G(\pi/4) - G(\pi/16)]$

(C) $\pi[G(1/2) - G(1/4)]$ (D) $G(1/\sqrt{2}) - G(1/2)$

Open solution

Q223. AIEEE 2012 (Online) 7th May

If $\int_e^x tf(t)\, dt = \sin x - x\cos x - \frac{x^2}{2}$, for all $x \in R - \{0\}$, then the value of $f\left(\frac{\pi}{6}\right)$ is

(A) $1/2$ (B) 1 (C) 0 (D) $-1/2$

Open solution

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Q224. AIEEE 2011

The value of $\int_0^1 \frac{8\log(1+x)}{1+x^2}dx$ is

(A) $\frac{\pi}{8}\log 2$ (B) $\frac{\pi}{2}\log 2$ (C) $\log 2$ (D) $\pi \log 2$

Open solution

Q225. AIEEE 2010

Let $p(x)$ be a function defined on R such that $p'(x) = p'(1-x)$, for all $x \in [0,1], p(0) = 1$ and $p(1) = 41$. Then $\int_0^1 p(x)dx$ equals :

(A) 21 (B) 41 (C) 42 (D) $\sqrt{41}$

Open solution

Q226. AIEEE 2009

$\int_0^{\pi} [\cot x]dx$, where $[.]$ denotes the greatest integer function, is equal to:

(A) 1 (B) -1 (C) $-\frac{\pi}{2}$ (D) $\frac{\pi}{2}$

Open solution

Q227. AIEEE 2008

Let $I = \int_0^1 \frac{\sin x}{\sqrt{x}}dx$ and $J = \int_0^1 \frac{\cos x}{\sqrt{x}}dx$. Then which one of the following is true?

(A) $I > \frac{2}{3}$ and $J > 2$ (B) $I < \frac{2}{3}$ and $J < 2$
(C) $I < \frac{2}{3}$ and $J > 2$ (D) $I > \frac{2}{3}$ and $J < 2$

Open solution

Q228. AIEEE 2007

Let $F(x) = f(x) + f\left(\frac{1}{x}\right)$, where $f(x) = \int_1^x \frac{\log t}{1+t}dt$, Then $F(e)$ equals

(A) 1 (B) 2 (C) $1/2$ (D) 0

Open solution

Q229. AIEEE 2007

The solution for x of the equation $\int_{\sqrt{2}}^{x} \frac{dt}{t\sqrt{t^2-1}} = \frac{\pi}{2}$ is

(A) $\frac{\sqrt{3}}{2}$ (B) $2\sqrt{2}$ (C) 2 (D) None

Open solution

Q230. AIEEE 2006

The value of integral, $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}}dx$ is

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 2 (D) 1

Open solution

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Q231. AIEEE 2006

$\int_0^{\pi} x f(\sin x)dx$ is equal to

(A) $\pi \int_0^{\pi} f(\cos x)dx$

(B) $\pi \int_0^{\pi} f(sinx)dx$

(C) $\frac{\pi}{2} \int_0^{\pi/2} f(sinx)dx$

(D) $\pi \int_0^{\pi/2} f(\cos x)dx$

Open solution

Q232. AIEEE 2006

$\int_{-\frac{3\pi}{2}}^{-\frac{\pi}{2}} \left[(x+\pi)^3 + \cos^2(x+3\pi)\right]dx$ is equal to

(A) $\frac{\pi^4}{32}$

(B) $\frac{\pi^4}{32} + \frac{\pi}{2}$

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{4} - 1$

Open solution

Q233. AIEEE 2006

The value of $\int_1^a [x]f'(x)dx, a > 1$ where $[x]$ denotes the greatest integer not exceeding x is

(A) $af(a) - \{f(1) + f(2) + \ldots\ldots\ldots\ldots f([a])\}$

(B) $[a]f(a) - \{f(1) + f(2) + \ldots\ldots\ldots\ldots f([a])\}$

(C) $[a]f([a]) - \{f(1) + f(2) + \ldots\ldots\ldots\ldots f(a)\}$

(D) $af([a]) - \{f(1) + f(2) + \ldots\ldots\ldots\ldots f(a)\}$

Open solution

Q234. AIEEE 2005

Let $f : R \to R$ be a differentiable function having $f(2) = 6$, $f'(2) = \left(\frac{1}{48}\right)$. Then $\lim_{x \to 2} \int_6^{f(x)} \frac{4t^3}{x-2}dt$ equals :

(A) 24 (B) 36 (C) 12 (D) 18

Open solution

Q235. AIEEE 2005

If $I_1 = \int_0^1 2^{x^2}dx, I_2 = \int_0^1 2^{x^3}dx, I_3 = \int_1^2 2^{x^2}dx$ and $I_4 = \int_1^2 2^{x^3}dx$ then

(A) $I_2 > I_1$ (B) $I_1 > I_2$ (C) $I_3 = I_4$ (D) $I_3 > I_4$

Open solution

Q236. AIEEE 2005

The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x}dx,\ a > 0$, is

(A) $a\pi$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{a}$ (D) 2π

Open solution

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Q237. AIEEE 2004

The value of $\int_{-2}^{3}|1-x^2|dx$ is

(A) $\frac{1}{3}$ (B) $\frac{14}{3}$ (C) $\frac{7}{3}$ (D) $\frac{28}{3}$

Open solution

Q238. AIEEE 2004

The value of $I=\int_{0}^{\pi/2}\frac{(\sin x+\cos x)^2}{\sqrt{1+\sin 2x}}dx$ is

(A) 3 (B) 1 (C) 2 (D) 0

Open solution

Q239. AIEEE 2004

If $\int_{0}^{\pi}xf(\sin x)dx=A\int_{0}^{\pi/2}f(\sin x)dx$, then A is

(A) 2π (B) π (C) $\frac{\pi}{4}$ (D) 0

Open solution

Q240. AIEEE 2004

If $f(x)=\frac{e^x}{1+e^x}, I_1=\int_{f(-a)}^{f(a)}xg\{x(1-x)\}dx$

and $I_2=\int_{f(-a)}^{f(a)}g\{x(1-x)\}dx$, then the value of $\frac{I_2}{I_1}$ is

(A) 1 (B) -3 (C) -1 (D) 2

Open solution

Q241. AIEEE 2003

If $f(y)=e^y, g(y)=y; y>0$ and

$F(t)=\int_{0}^{t}f(t-y)g(y)dy$, then :

(A) $F(t)=te^{-t}$ (B) $F(t)=1t-te^{-1}(1+t)$

(C) $F(t)=e^t-(1+t)$ (D) $F(t)=te^t$.

Open solution

Q242. AIEEE 2003

If $f(a+b-x)=f(x)$ then $\int_{a}^{b}xf(x)dx$ is equal to

(A) $\frac{a+b}{2}\int_{a}^{b}f(a+b+x)dx$ (B) $\frac{a+b}{2}\int_{a}^{b}f(b-x)dx$

(C) $\frac{a+b}{2}\int_{a}^{b}f(x)dx$ (D) $\frac{b-a}{2}\int_{a}^{b}f(x)dx$

Open solution

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Q243. AIEEE 2003

Let $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and $g(x)$ be a function that satisfies $f(x) + g(x) = x^2$. Then the value of the integral $\int_0^1 f(x)g(x)dx$, is

(A) $e + \frac{e^2}{2} + \frac{5}{2}$ (B) $e - \frac{e^2}{2} - \frac{5}{2}$

(C) $e + \frac{e^2}{2} - \frac{3}{2}$ (D) $e - \frac{e^2}{2} - \frac{3}{2}$

Open solution

Q244. AIEEE 2003

The value of the integral $I = \int_0^1 x(1-x)^n dx$ is

(A) $\frac{1}{n+1} + \frac{1}{n+2}$ (B) $\frac{1}{n+1}$

(C) $\frac{1}{n+2}$ (D) $\frac{1}{n+1} - \frac{1}{n+2}$

Open solution

Q245. AIEEE 2003

The value of $\lim_{x\to 0} \frac{\int_0^{x^2} \sec^2 t dt}{x \sin x}$ is

(A) 0 (B) 3 (C) 2 (D) 1

Open solution

Q246. AIEEE 2002

$\int_0^{10\pi} |\sin x| dx$ is

(A) 20 (B) 8 (C) 10 (D) 18

Open solution

Q247. AIEEE 2002

$I_n = \int_0^{\pi/4} \tan^n x \, dx$ then $\lim_{n\to\infty} n\left[I_n + I_{n+2}\right]$ equals

(A) $\frac{1}{2}$ (B) 1 (C) ∞ (D) zero

Open solution

Q248. AIEEE 2002

$\int_0^2 \left[x^2\right] dx$ is

(A) $2 - \sqrt{2}$ (B) $2 + \sqrt{2}$ (C) $\sqrt{2} - 1$ (D) $-\sqrt{2} - \sqrt{3} + 5$

Open solution

Q249. AIEEE 2002

$\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x} dx$ is

(A) $\frac{\pi^2}{4}$ (B) π^2 (C) zero (D) $\frac{\pi}{2}$

Open solution

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Q250. AIEEE 2002

If $y = f(x)$ makes +ve intercept of 2 and 0 unit on x and y axes and encloses an area of $3/4$ square unit with the axes then $\int_0^2 x f'(x)dx$ is

(A) $3/2$ (B) 1 (C) $5/4$ (D) $-3/4$

Open solution

Numerical 77 questions

Q251. JEE Main 2026 (Online) 8th April Evening Shift

If $\int_{\pi/6}^{\pi/4} \left(\cot\left(x-\frac{\pi}{3}\right)\cot\left(x+\frac{\pi}{3}\right)+1\right)dx = \alpha \log_e(\sqrt{3}-1)$, then $9\alpha^2$ is equal to ____.

Open solution

Q252. JEE Main 2026 (Online) 2nd April Morning Shift

If $\alpha = \int_0^{2\sqrt{3}} \log_2(x^2+4)\,dx + \int_2^4 \sqrt{2^x-4}\,dx$, then α^2 is equal to _____.

Open solution

Q253. JEE Main 2026 (Online) 28th January Evening Shift

Let f be a differentiable function satisfying $f(x) = 1 - 2x + \int_0^x e^{(x-t)} f(t)\,dt, x \in \mathbb{R}$ and let

$g(x) = \int_0^x (f(t)+2)^{15}(t-4)^6(t+12)^{17}\,dt, x \in \mathbb{R}$.

If p and q are respectively the points of local minima and local maxima of g, then the value of $|p+q|$ is equal to _____.

Open solution

Q254. JEE Main 2026 (Online) 28th January Morning Shift

The value of $\sum_{r=1}^{20} \left(\left|\sqrt{\pi\left(\int_0^r x|\sin \pi x|dx\right)}\right|\right)$ is ____

Open solution

Q255. JEE Main 2026 (Online) 24th January Evening Shift

If $f(x)$ satisfies the relation $f(x) = e^x + \int_0^1 (y + xe^x) f(y)dy$, then $e + f(0)$ is equal to ____.

Open solution

Q256. JEE Main 2026 (Online) 24th January Morning Shift

Let a differentiable function f satisfy the equation $\int_0^{36} f\left(\frac{tx}{36}\right)dt = 4\alpha f(x)$. If $y = f(x)$ is a standard parabola passing through the points $(2,1)$ and $(-4,\beta)$, then β^α is equal to ____.

Open solution

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Q257. JEE Main 2026 (Online) 23rd January Evening Shift

The number of elements in the set $S = \{x : x \in [0, 100] \text{ and } \int_0^x t^2 \sin(x-t)dt = x^2\}$ is ____

Open solution

Q258. JEE Main 2026 (Online) 22nd January Evening Shift

Let [.] be the greatest integer function. If $\alpha = \int_0^{64} \left(x^{1/3} - \left[x^{1/3}\right]\right)dx$, then $\frac{1}{\pi}\int_0^{\alpha\pi} \left(\frac{\sin^2\theta}{\sin^6\theta+\cos^6\theta}\right)d\theta$ is equal to ____ .

Open solution

Q259. JEE Main 2026 (Online) 21st January Evening Shift

Let $[\cdot]$ denote the greatest integer function and $f(x) = \lim_{n\to\infty} \frac{1}{n^3}\sum_{k=1}^{n}\left[\frac{k^2}{3^x}\right]$. Then $12\sum_{j=1}^{\infty} f(j)$ is equal to _____.

Open solution

Q260. JEE Main 2026 (Online) 21st January Evening Shift

If $\int_0^1 4\cot^{-1}\left(1-2x+4x^2\right)dx = a\tan^{-1}(2) - b\log_e(5)$, where $a, b \in N$, then $(2a+b)$ is equal to ____ .

Open solution

Q261. JEE Main 2026 (Online) 21st January Morning Shift

$6\int_0^{\pi} |(\sin 3x + \sin 2x + \sin x)|dx$ is equal to ____ .

Open solution

Q262. JEE Main 2025 (Online) 2nd April Morning Shift

Let [.] denote the greatest integer function. If $\int_0^{e^3} \left[\frac{1}{e^{x-1}}\right]dx = \alpha - \log_e 2$, then α^3 is equal to _____.

Open solution

Q263. JEE Main 2025 (Online) 29th January Evening Shift

If $24\int_0^{\frac{\pi}{4}} \left(\sin\left|4x - \frac{\pi}{12}\right| + [2\sin x]\right)dx = 2\pi + \alpha$, where $[\cdot]$ denotes the greatest integer function, then α is equal to _____.

Open solution

Q264. JEE Main 2025 (Online) 29th January Evening Shift

If $\lim_{t\to 0}\left(\int_0^1 (3x+5)^t dx\right)^{\frac{1}{t}} = \frac{\alpha}{5e}\left(\frac{8}{5}\right)^{\frac{2}{3}}$, then α is equal to _____.

Open solution

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Q265. JEE Main 2025 (Online) 29th January Morning Shift

Let $f:(0,\infty)\to\mathbf{R}$ be a twice differentiable function. If for some $a\neq 0, \int_0^1 f(\lambda x)\mathrm{d}\lambda = af(x), f(1)=1$ and $f(16)=\frac{1}{8}$, then $16-f'\left(\frac{1}{16}\right)$ is equal to _____.

Open solution

Q266. JEE Main 2024 (Online) 9th April Morning Shift

Let $\lim\limits_{n\to\infty}\left(\frac{n}{\sqrt{n^4+1}}-\frac{2n}{(n^2+1)\sqrt{n^4+1}}+\frac{n}{\sqrt{n^4+16}}-\frac{8n}{(n^2+4)\sqrt{n^4+16}}+\ldots+\frac{n}{\sqrt{n^4+n^4}}-\frac{2n\cdot n^2}{(n^2+n^2)\sqrt{n^4+n^4}}\right)$ be $\frac{\pi}{k}$, using only the principal values of the inverse trigonometric functions. Then k^2 is equal to _____.

Open solution

Q267. JEE Main 2024 (Online) 6th April Evening Shift

Let $[t]$ denote the largest integer less than or equal to t. If $\int_0^3\left([x^2]+\left[\frac{x^2}{2}\right]\right)\mathrm{d}x=\mathrm{a}+\mathrm{b}\sqrt{2}-\sqrt{3}-\sqrt{5}+\mathrm{c}\sqrt{6}-\sqrt{7}$, where $\mathrm{a},\mathrm{b},\mathrm{c}\in\mathbf{Z}$, then $\mathrm{a}+\mathrm{b}+\mathrm{c}$ is equal to _____.

Open solution

Q268. JEE Main 2024 (Online) 6th April Morning Shift

Let $r_k=\frac{\int_0^1(1-x^7)^k dx}{\int_0^1(1-x^7)^{k+1}dx}, k\in\mathbb{N}$. Then the value of $\sum_{k=1}^{10}\frac{1}{7(r_k-1)}$ is equal to _____.

Open solution

Q269. JEE Main 2024 (Online) 5th April Evening Shift

If $f(t)=\int_0^{\pi}\frac{2x\,\mathrm{d}x}{1-\cos^2\mathrm{t}\sin^2 x}, 0<\mathrm{t}<\pi$, then the value of $\int_0^{\frac{\pi}{2}}\frac{\pi^2\mathrm{dt}}{f(\mathrm{t})}$ equals _____.

Open solution

Q270. JEE Main 2024 (Online) 4th April Morning Shift

If the shortest distance between the lines $\frac{x+2}{2}=\frac{y+3}{3}=\frac{z-5}{4}$ and $\frac{x-3}{1}=\frac{y-2}{-3}=\frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}\mathrm{k}$, and $\int_0^{\mathrm{k}}[x^2]\mathrm{d}x=\alpha-\sqrt{\alpha}$, where $[x]$ denotes the greatest integer function, then $6\alpha^3$ is equal to _____.

Open solution

Q271. JEE Main 2024 (Online) 4th April Morning Shift

If $\int_0^{\frac{\pi}{4}}\frac{\sin^2 x}{1+\sin x\cos x}\,\mathrm{d}x=\frac{1}{\mathrm{a}}\log_{\mathrm{e}}\left(\frac{\mathrm{a}}{3}\right)+\frac{\pi}{\mathrm{b}\sqrt{3}}$, where $\mathrm{a},\mathrm{b}\in\mathrm{N}$, then $\mathrm{a}+\mathrm{b}$ is equal to _____.

Open solution

Q272. JEE Main 2024 (Online) 1st February Evening Shift

Let $f:(0,\infty)\to\mathbf{R}$ and $\mathrm{F}(x)=\int_0^x \mathrm{t}f(\mathrm{t})\mathrm{dt}$. If $\mathrm{F}\left(x^2\right)=x^4+x^5$, then $\sum_{\mathrm{r}=1}^{12}f\left(\mathrm{r}^2\right)$ is equal to _____.

Open solution

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Q273. JEE Main 2024 (Online) 1st February Morning Shift

If $\int_{-\pi/2}^{\pi/2} \frac{8\sqrt{2}\cos x\,dx}{(1+e^{\sin x})(1+\sin^4 x)} = \alpha\pi + \beta\log_e(3+2\sqrt{2})$, where α, β are integers, then $\alpha^2+\beta^2$ equals :

Open solution

Q274. JEE Main 2024 (Online) 31st January Evening Shift

$\left|\frac{120}{\pi^3}\int_0^{\pi} \frac{x^2\sin x\cos x}{\sin^4 x+\cos^4 x}dx\right|$ is equal to _____.

Open solution

Q275. JEE Main 2024 (Online) 31st January Morning Shift

If the integral $525\int_0^{\frac{\pi}{2}} \sin 2x\cos^{\frac{11}{2}} x\left(1+\mathrm{Cos}^{\frac{5}{2}} x\right)^{\frac{1}{2}}dx$ is equal to $(n\sqrt{2}-64)$, then n is equal to _____.

Open solution

Q276. JEE Main 2024 (Online) 31st January Morning Shift

Let $S=(-1,\infty)$ and $f: S\to\mathbb{R}$ be defined as

$$f(x)=\int_{-1}^{x}\left(e^t-1\right)^{11}(2t-1)^5(t-2)^7(t-3)^{12}(2t-10)^{61}dt,$$

Let $\mathrm{p}=$ Sum of squares of the values of x, where $f(x)$ attains local maxima on S, and $\mathrm{q}=$ Sum of the values of x, where $f(x)$ attains local minima on S. Then, the value of p^2+2q is _____.

Open solution

Q277. JEE Main 2024 (Online) 31st January Morning Shift

Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=\frac{4^x}{4^x+2}$ and $M=\int_{f(a)}^{f(1-a)} x\sin^4(x(1-x))dx, N=\int_{f(a)}^{f(1-a)}\sin^4(x(1-x))dx; a\neq\frac{1}{2}$. If $\alpha M=\beta N, \alpha,\beta\in\mathbb{N}$, then the least value of $\alpha^2+\beta^2$ is equal to _____.

Open solution

Q278. JEE Main 2024 (Online) 30th January Morning Shift

The value of $9\int_0^9\left[\sqrt{\frac{10x}{x+1}}\right]\mathrm{d}x$, where $[t]$ denotes the greatest integer less than or equal to t, is

Open solution

Q279. JEE Main 2024 (Online) 29th January Evening Shift

Let the slope of the line $45x+5y+3=0$ be $27r_1+\frac{9r_2}{2}$ for some $r_1,r_2\in\mathbb{R}$. Then $\lim_{x\to 3}\left(\int_3^x \frac{8t^2}{\frac{3r_2x}{2}-r_2x^2-r_1x^3-3x}dt\right)$ is equal to _____.

Open solution

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Q280. JEE Main 2024 (Online) 29th January Evening Shift

If $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sqrt{1-\sin 2x}dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3}$, where α, β and γ are rational numbers, then $3\alpha + 4\beta - \gamma$ is equal to _____.

Open solution

Q281. JEE Main 2024 (Online) 27th January Evening Shift

Let $f(x) = \int_0^x g(t)\log_e\left(\frac{1-t}{1+t}\right)dt$, where g is a continuous odd function. If $\int_{-\pi/2}^{\pi/2}\left(f(x) + \frac{x^2\cos x}{1+e^x}\right)dx = \left(\frac{\pi}{\alpha}\right)^2 - \alpha$, then α is equal to _____.

Open solution

Q282. JEE Main 2023 (Online) 13th April Evening Shift

Let $f_n = \int_0^{\frac{\pi}{2}}\left(\sum_{k=1}^{n}\sin^{k-1}x\right)\left(\sum_{k=1}^{n}(2k-1)\sin^{k-1}x\right)\cos x dx, n \in \mathbb{N}$. Then $f_{21} - f_{20}$ is equal to _____

Open solution

Q283. JEE Main 2023 (Online) 13th April Morning Shift

Let for $x \in \mathbb{R}, S_0(x) = x, S_k(x) = C_k x + k\int_0^x S_{k-1}(t)dt$, where

$C_0 = 1, C_k = 1 - \int_0^1 S_{k-1}(x)dx, k = 1, 2, 3, \ldots$ Then $S_2(3) + 6C_3$ is equal to _____.

Open solution

Q284. JEE Main 2023 (Online) 12th April Morning Shift

If $\int_{-0.15}^{0.15}|100x^2 - 1|dx = \frac{k}{3000}$, then k is equal to _____.

Open solution

Q285. JEE Main 2023 (Online) 11th April Morning Shift

For $m, n > 0$, let $\alpha(m,n) = \int_0^2 t^m(1+3t)^n dt$. If $11\alpha(10,6) + 18\alpha(11,5) = p(14)^6$, then p is equal to _____.

Open solution

Q286. JEE Main 2023 (Online) 8th April Evening Shift

Let $[t]$ denote the greatest integer function. If $\int_0^{2.4}\left[x^2\right]dx = \alpha + \beta\sqrt{2} + \gamma\sqrt{3} + \delta\sqrt{5}$, then $\alpha + \beta + \gamma + \delta$ is equal to _____.

Open solution

Q287. JEE Main 2023 (Online) 8th April Morning Shift

Let $[t]$ denote the greatest integer $\leq t$. Then $\frac{2}{\pi}\int_{\pi/6}^{5\pi/6}(8[\operatorname{cosec} x] - 5[\cot x])dx$ is equal to _____.

Open solution

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Q288. JEE Main 2023 (Online) 1st February Evening Shift

If $\int_0^{\pi} \frac{5^{\cos x}(1+\cos x \cos 3x+\cos^2 x+\cos^3 x \cos 3x)dx}{1+5^{\cos x}} = \frac{k\pi}{16}$, then k is equal to _____.

Open solution

Q289. JEE Main 2023 (Online) 1st February Morning Shift

If $\int_0^1 \left(x^{21}+x^{14}+x^7\right)\left(2x^{14}+3x^7+6\right)^{1/7}dx = \frac{1}{l}(11)^{m/n}$ where $l, m, n \in \mathbb{N}$, m and n are coprime then $l+m+n$ is equal to _____.

Open solution

Q290. JEE Main 2023 (Online) 1st February Morning Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $f'(x) + f(x) = \int_0^2 f(t)dt$. If $f(0) = e^{-2}$, then $2f(0) - f(2)$ is equal to _____.

Open solution

Q291. JEE Main 2023 (Online) 30th January Morning Shift

$\lim_{x\to 0} \frac{48}{x^4} \int_0^x \frac{t^3}{t^6+1}\,dt$ is equal to _____.

Open solution

Q292. JEE Main 2023 (Online) 25th January Evening Shift

If $\int_{\frac{1}{3}}^{3} |\log_e x|dx = \frac{m}{n}\log_e\left(\frac{n^2}{e}\right)$, where m and n are coprime natural numbers, then m^2+n^2-5 is equal to _____.

Open solution

Q293. JEE Main 2023 (Online) 24th January Evening Shift

Let f be a differentiable function defined on $\left[0, \frac{\pi}{2}\right]$ such that $f(x) > 0$ and $f(x) + \int_0^x f(t)\sqrt{1-(\log_e f(t))^2}dt = e, \forall x \in \left[0, \frac{\pi}{2}\right]$. Then $\left(6\log_e f\left(\frac{\pi}{6}\right)\right)^2$ is equal to _____.

Open solution

Q294. JEE Main 2023 (Online) 24th January Morning Shift

The value of $12\int_0^3 \left|x^2-3x+2\right|dx$ is _____

Open solution

Q295. JEE Main 2023 (Online) 24th January Morning Shift

The value of $\frac{8}{\pi}\int_0^{\frac{\pi}{2}} \frac{(\cos x)^{2023}}{(\sin x)^{2023}+(\cos x)^{2023}}dx$ is _____

Open solution

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Q296. JEE Main 2022 (Online) 28th July Evening Shift

The value of the integral $\int_0^{\frac{\pi}{2}} 60\frac{\sin(6x)}{\sin x}dx$ is equal to _____.

Open solution

Q297. JEE Main 2022 (Online) 28th July Morning Shift

If $\int_0^{\sqrt{3}} \frac{15x^3}{\sqrt{1+x^2+\sqrt{(1+x^2)^3}}}\,\mathrm{d}x = \alpha\sqrt{2}+\beta\sqrt{3}$, where α, β are integers, then $\alpha+\beta$ is equal to _____.

Open solution

Q298. JEE Main 2022 (Online) 27th July Evening Shift

Let $f(x) = \min\{[x-1],[x-2],\ldots,[x-10]\}$ where [t] denotes the greatest integer $\leq$ t. Then $\int_0^{10} f(x)\mathrm{d}x + \int_0^{10}(f(x))^2\,\mathrm{d}x + \int_0^{10}|f(x)|\mathrm{d}x$ is equal to _____.

Open solution

Q299. JEE Main 2022 (Online) 27th July Evening Shift

Let f be a differentiable function satisfying $f(x) = \frac{2}{\sqrt{3}}\int_0^{\sqrt{3}} f\left(\frac{\lambda^2 x}{3}\right)\mathrm{d}\lambda, x>0$ and $f(1)=\sqrt{3}$. If $y=f(x)$ passes through the point $(\alpha, 6)$, then α is equal to _____.

Open solution

Q300. JEE Main 2022 (Online) 26th July Morning Shift

If $\mathrm{n}(2\mathrm{n}+1)\int_0^1(1-x^{\mathrm{n}})^{2\mathrm{n}}\mathrm{d}x = 1177\int_0^1(1-x^{\mathrm{n}})^{2\mathrm{n}+1}\,\mathrm{d}x$, then $\mathrm{n}\in\mathbf{N}$ is equal to _____.

Open solution

Q301. JEE Main 2022 (Online) 25th July Evening Shift

Let f be a twice differentiable function on $\mathbb{R}$. If $f'(0)=4$ and $f(x)+\int_0^x(x-t)f'(t)dt = \left(e^{2x}+e^{-2x}\right)\cos 2x + \frac{2}{a}x$, then $(2a+1)^5a^2$ is equal to _____.

Open solution

Q302. JEE Main 2022 (Online) 25th July Evening Shift

Let $a_n = \int_{-1}^{n}\left(1+\frac{x}{2}+\frac{x^2}{3}+\ldots..+\frac{x^{n-1}}{n}\right)dx$ for every n $\in$ N. Then the sum of all the elements of the set {n $\in$ N : $a_n \in (2, 30)$} is _____.

Open solution

Q303. JEE Main 2022 (Online) 30th June Morning Shift

Let $f(t) = \int_0^t e^{x^3}\left(\frac{x^8}{(x^6+2x^3+2)^2}\right)dx$. If $f(1)+f'(1) = \alpha e - \frac{1}{6}$, then the value of 150α is equal to _____.

Open solution

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Q304. JEE Main 2022 (Online) 26th June Evening Shift

The integral $\frac{24}{\pi}\int_0^{\sqrt{2}}\frac{(2-x^2)dx}{(2+x^2)\sqrt{4+x^4}}$ is equal to _____.

Open solution

Q305. JEE Main 2022 (Online) 26th June Morning Shift

Let f(x) = max {|x + 1|, |x + 2|, ..., |x + 5|}. Then $\int_{-6}^{0} f(x)dx$ is equal to _____.

Open solution

Q306. JEE Main 2022 (Online) 26th June Morning Shift

The value of the integral

$\frac{48}{\pi^4}\int_0^{\pi}\left(\frac{3\pi x^2}{2}-x^3\right)\frac{\sin x}{1+\cos^2 x}dx$ is equal to _____.

Open solution

Q307. JEE Main 2022 (Online) 25th June Evening Shift

The value of b > 3 for which $12\int_3^b\frac{1}{(x^2-1)(x^2-4)}dx=\log_e\left(\frac{49}{40}\right)$, is equal to _____.

Open solution

Q308. JEE Main 2022 (Online) 24th June Morning Shift

Let $f(\theta)=\sin\theta+\int_{-\pi/2}^{\pi/2}(\sin\theta+t\cos\theta)f(t)dt$. Then the value of $\left|\int_0^{\pi/2}f(\theta)d\theta\right|$ is _____.

Open solution

Q309. JEE Main 2022 (Online) 24th June Morning Shift

Let $\underset{0\le x\le 2}{Max}\left\{\frac{9-x^2}{5-x}\right\}=\alpha$ and $\underset{0\le x\le 2}{Min}\left\{\frac{9-x^2}{5-x}\right\}=\beta$.

If $\int_{\beta-\frac{8}{3}}^{2\alpha-1}Max\left\{\frac{9-x^2}{5-x},x\right\}dx=\alpha_1+\alpha_2\log_e\left(\frac{8}{15}\right)$ then $\alpha_1+\alpha_2$ is equal to _____.

Open solution

Q310. JEE Main 2021 (Online) 31st August Morning Shift

Let [t] denote the greatest integer $\le$ t. Then the value of

8. $\int_{-\frac{1}{2}}^{1}([2x]+|x|)dx$ is _____.

Open solution

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Q311. JEE Main 2021 (Online) 31st August Morning Shift

If $x\phi(x) = \int_5^x (3t^2 - 2\phi'(t))dt$, x > -2, and $\phi(0) = 4$, then $\phi(2)$ is _____.

Open solution

Q312. JEE Main 2021 (Online) 27th July Evening Shift

If $\int_0^{\pi} (\sin^3 x)e^{-\sin^2 x}dx = \alpha - \frac{\beta}{e}\int_0^1 \sqrt{t}e^t dt$, then $\alpha + \beta$ is equal to _____.

Open solution

Q313. JEE Main 2021 (Online) 27th July Morning Shift

Let the domain of the function

$f(x) = \log_4\left(\log_5\left(\log_3(18x - x^2 - 77)\right)\right)$ be (a, b). Then the value of the integral $\int_a^b \frac{\sin^3 x}{\sin^3 x + \sin^3(a+b-x)}dx$ is equal to _____.

Open solution

Q314. JEE Main 2021 (Online) 27th July Morning Shift

Let $F : [3, 5] \to R$ be a twice differentiable function on (3, 5) such that

$F(x) = e^{-x}\int_3^x (3t^2 + 2t + 4F'(t))dt$. If $F'(4) = \frac{\alpha e^{\beta} - 224}{(e^{\beta} - 4)^2}$, then $\alpha + \beta$ is equal to _____.

Open solution

Q315. JEE Main 2021 (Online) 18th March Evening Shift

Let P(x) be a real polynomial of degree 3 which vanishes at x = -3. Let P(x) have local minima at x = 1, local maxima at x = -1 and $\int_{-1}^{1} P(x)dx$ = 18, then the sum of all the coefficients of the polynomial P(x) is equal to _____.

Open solution

Q316. JEE Main 2021 (Online) 18th March Morning Shift

Let f(x) and g(x) be two functions satisfying $f(x^2) + g(4 - x) = 4x^3$ and $g(4 - x) + g(x) = 0$, then the value of $\int_{-4}^{4} f(x^2)dx$ is

Open solution

Q317. JEE Main 2021 (Online) 17th March Evening Shift

Let $I_n = \int_1^e x^{19}(\log|x|)^n dx$, where n∈N. If $(20)I_{10} = \alpha I_9 + \beta I_8$, for natural numbers α and β, then $\alpha - \beta$ equals to _____.

Open solution

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Q318. JEE Main 2021 (Online) 17th March Morning Shift

If [.] represents the greatest integer function, then the value of

$\left|\int_0^{\sqrt{\frac{\pi}{2}}} \left[[x^2] - \cos x\right] dx\right|$ is _____.

Open solution

Q319. JEE Main 2021 (Online) 16th March Morning Shift

Let f : R $\rightarrow$ R be a continuous function such that f(x) + f(x + 1) = 2, for all x$\in$R.

If $I_1 = \int_0^8 f(x)dx$ and $I_2 = \int_{-1}^3 f(x)dx$, then the value of $I_1 + 2I_2$ is equal to _____.

Open solution

Q320. JEE Main 2021 (Online) 16th March Morning Shift

If the normal to the curve y(x) = $\int_0^x (2t^2 - 15t + 10)dt$ at a point (a, b) is parallel to the line x + 3y = -5, a > 1, then the value of | a + 6b | is equal to _____.

Open solution

Q321. JEE Main 2021 (Online) 26th February Evening Shift

If $I_{m,n} = \int_0^1 x^{m-1}(1-x)^{n-1}dx$, for m, $n \geq 1$, and

$\int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}dx = \alpha I_{m,n} \alpha \in R$, then α equals _____.

Open solution

Q322. JEE Main 2021 (Online) 26th February Morning Shift

The value of the integral $\int_0^{\pi} |\sin 2x| dx$ is _____.

Open solution

Q323. JEE Main 2021 (Online) 25th February Evening Shift

The value of $\int_{-2}^{2} |3x^2 - 3x - 6| dx$ is _____.

Open solution

Q324. JEE Main 2021 (Online) 24th February Morning Shift

If $\int_{-a}^{a} (|x| + |x-2|)dx = 22$, (a > 2) and [x] denotes the greatest integer $\leq$ x, then $\int_a^{-a} (x + [x])dx$ is equal to _____.

Open solution

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Q325. JEE Main 2020 (Online) 4th September Evening Slot

Let {x} and [x] denote the fractional part of x and the greatest integer $\leq$ x respectively of a real number x. If $\int_0^n \{x\}dx$, $\int_0^n [x]dx$ and 10(n^2 – n), $(n \in N, n > 1)$ are three consecutive terms of a G.P., then n is equal to_____.

Open solution

Q326. JEE Main 2020 (Online) 2nd September Evening Slot

Let [t] denote the greatest integer less than or equal to t.

Then the value of $\int_1^2 |2x - [3x]|dx$ is _____.

Open solution

Q327. JEE Main 2020 (Online) 2nd September Morning Slot

The integral $\int_0^2 ||x-1| - x|dx$

is equal to_____.

Open solution

MCQ (More than One Correct Answer) 1 questions

Q328. AIEEE 2012

If $g(x) = \int_0^x \cos 4t\, dt$, then $g(x+\pi)$ equals

(A) $\frac{g(x)}{g(\pi)}$

(B) $g(x) + g(\pi)$

(C) $g(x) - g(\pi)$

(D) $g(x).g(\pi)$

Open solution

Out of syllabus 23 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q329. JEE Main 2023 (Online) 13th April Morning Shift · MCQ (Single Correct Answer)

Among

(S1): $\lim_{n\to\infty} \frac{1}{n^2}(2+4+6+\ldots\ldots+2n) = 1$

(S2): $\lim_{n\to\infty} \frac{1}{n^{16}}\left(1^{15}+2^{15}+3^{15}+\ldots\ldots+n^{15}\right) = \frac{1}{16}$

(A) Only (S1) is true

(B) Both (S1) and (S2) are true

(C) Both (S1) and (S2) are false

(D) Only (S2) is true

Open solution

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Q330. JEE Main 2023 (Online) 1st February Morning Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left[\frac{1}{1+n}+\frac{1}{2+n}+\frac{1}{3+n}+\ldots+\frac{1}{2n}\right]$ is equal to

(A) 0
(B) $\log_e 2$
(C) $\log_e\left(\frac{2}{3}\right)$
(D) $\log_e\left(\frac{3}{2}\right)$

Open solution

Q331. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\frac{3}{n}\left\{4+\left(2+\frac{1}{n}\right)^2+\left(2+\frac{2}{n}\right)^2+\ldots+\left(3-\frac{1}{n}\right)^2\right\}$ is equal to :

(A) 0
(B) $\frac{19}{3}$
(C) 19
(D) 12

Open solution

Q332. JEE Main 2022 (Online) 30th July Evening Shift · MCQ (Single Correct Answer)

$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n\left(n^2+k^2\right)}{\left(n^4+n^2k^2+k^4\right)}$
is equal to

(A) $\frac{\pi}{\sqrt{3}}$
(B) $\frac{2\pi}{3\sqrt{3}}$
(C) $\frac{\pi}{2\sqrt{3}}$
(D) $\frac{\pi}{3}$

Open solution

Q333. JEE Main 2022 (Online) 26th July Morning Shift · MCQ (Single Correct Answer)

If $a=\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\frac{2n}{n^2+k^2}$ and $f(x)=\sqrt{\frac{1-\cos x}{1+\cos x}}$, $x\in(0,1)$, then :

(A) $2\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$
(B) $f\left(\frac{a}{2}\right)f'\left(\frac{a}{2}\right)=\sqrt{2}$
(C) $\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$
(D) $f\left(\frac{a}{2}\right)=\sqrt{2}f'\left(\frac{a}{2}\right)$

Open solution

Q334. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\frac{1}{2^n}\left(\frac{1}{\sqrt{1-\frac{1}{2^n}}}+\frac{1}{\sqrt{1-\frac{2}{2^n}}}+\frac{1}{\sqrt{1-\frac{3}{2^n}}}+\ldots+\frac{1}{\sqrt{1-\frac{2^n-1}{2^n}}}\right)$ is equal to

(A) $\frac{1}{2}$
(B) 1
(C) 2
(D) -2

Open solution

Q335. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{r}{2r^2-7rn+6n^2}$ is equal to :

(A) $\log_e\left(\frac{\sqrt{3}}{2}\right)$
(B) $\log_e\left(\frac{3\sqrt{3}}{4}\right)$
(C) $\log_e\left(\frac{27}{4}\right)$
(D) $\log_e\left(\frac{4}{3}\right)$

Open solution

Q336. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left(\frac{n^2}{(n^2+1)(n+1)}+\frac{n^2}{(n^2+4)(n+2)}+\frac{n^2}{(n^2+9)(n+3)}+\ldots.+\frac{n^2}{(n^2+n^2)(n+n)}\right)$ is equal to :

(A) $\frac{\pi}{8}+\frac{1}{4}\log_e 2$
(B) $\frac{\pi}{4}+\frac{1}{8}\log_e 2$
(C) $\frac{\pi}{4}-\frac{1}{8}\log_e 2$
(D) $\frac{\pi}{8}+\log_e\sqrt{2}$

Open solution

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Q337. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

If $U_n = \left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)^2 \ldots\ldots \left(1+\frac{n^2}{n^2}\right)^n$, then $\lim\limits_{n\to\infty} (U_n)^{\frac{-4}{n^2}}$ is equal to :

(A) $\frac{e^2}{16}$ (B) $\frac{4}{e}$ (C) $\frac{16}{e^2}$ (D) $\frac{4}{e^2}$

Open solution

Q338. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

The value of

$\lim\limits_{n\to\infty} \frac{1}{n} \sum\limits_{r=0}^{2n-1} \frac{n^2}{n^2+4r^2}$ is :

(A) $\frac{1}{2}\tan^{-1}(2)$ (B) $\frac{1}{2}\tan^{-1}(4)$ (C) $\tan^{-1}(4)$ (D) $\frac{1}{4}\tan^{-1}(4)$

Open solution

Q339. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

The value of $\lim\limits_{n\to\infty} \frac{1}{n} \sum\limits_{j=1}^{n} \frac{(2j-1)+8n}{(2j-1)+4n}$ is equal to :

(A) $5+\log_e\left(\frac{3}{2}\right)$ (B) $2-\log_e\left(\frac{2}{3}\right)$

(C) $3+2\log_e\left(\frac{2}{3}\right)$ (D) $1+2\log_e\left(\frac{3}{2}\right)$

Open solution

Q340. JEE Main 2021 (Online) 25th February Evening Shift · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left[\frac{1}{n}+\frac{n}{(n+1)^2}+\frac{n}{(n+2)^2}+\ldots\ldots\ldots+\frac{n}{(2n-1)^2}\right]$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) 1 (D) $\frac{1}{4}$

Open solution

Q341. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left(\frac{(n+1)^{1/3}}{n^{4/3}}+\frac{(n+2)^{1/3}}{n^{4/3}}+\ldots\ldots\ldots+\frac{(2n)^{1/3}}{n^{4/3}}\right)$ is equal to :

(A) $\frac{4}{3}(2)^{3/4}$ (B) $\frac{3}{4}(2)^{4/3}-\frac{3}{4}$

(C) $\frac{4}{3}(2)^{4/3}$ (D) $\frac{3}{4}(2)^{4/3}-\frac{4}{3}$

Open solution

Q342. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left(\frac{n}{n^2+1^2}+\frac{n}{n^2+2^2}+\frac{n}{n^2+3^2}+\ldots\ldots+\frac{1}{5n}\right)$ is equal to :

(A) $\tan^{-1}(2)$ (B) $\tan^{-1}(3)$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Open solution

Q343. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If $\lim\limits_{n\to\infty} \frac{1^a+2^a+\ldots\ldots+n^a}{(n+1)^{a-1}[(na+1)+(na+2)+\ldots\ldots+(na+n)]} = \frac{1}{60}$

for some positive real number a, then a is equal to :

(A) 7 (B) 8 (C) $\frac{15}{2}$ (D) $\frac{17}{2}$

Open solution

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Q344. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left(\frac{(n+1)(n+2)...3n}{n^{2n}}\right)^{\frac{1}{n}}$ is equal to:

(A) $\frac{9}{e^2}$ (B) $3\log 3-2$ (C) $\frac{18}{e^4}$ (D) $\frac{27}{e^2}$

Open solution

Q345. AIEEE 2005 · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2}+\frac{2}{n^2}\sec^2\frac{4}{n^2}....+\frac{1}{n}\sec^2 1\right]$ equals

(A) $\frac{1}{2}\sec 1$ (B) $\frac{1}{2}$cosec 1 (C) tan 1 (D) $\frac{1}{2}$tan 1

Open solution

Q346. AIEEE 2004 · MCQ (Single Correct Answer)

$Lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{1}{n}e^{\frac{r}{n}}$ is

(A) $e+1$ (B) $e-1$ (C) $1-e$ (D) e

Open solution

Q347. AIEEE 2003 · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\frac{1+2^4+3^4+....+n^4}{n^5}$ - $\lim\limits_{n\to\infty}\frac{1+2^3+3^3+....+n^3}{n^5}$

(A) $\frac{1}{5}$ (B) $\frac{1}{30}$ (C) zero (D) $\frac{1}{4}$

Open solution

Q348. AIEEE 2002 · MCQ (Single Correct Answer)

$\lim\limits_{n\to\infty}\frac{1^p+2^p+3^p+.....+n^p}{n^{p+1}}$ is

(A) $\frac{1}{p+1}$ (B) $\frac{1}{1-p}$ (C) $\frac{1}{p}-\frac{1}{p-1}$ (D) $\frac{1}{p+2}$

Open solution

Q349. JEE Main 2023 (Online) 6th April Evening Shift · Numerical

Let $f(x)=\frac{x}{(1+x^n)^{\frac{1}{n}}}, x\in\mathbb{R}-\{-1\}, n\in\mathbb{N}, n>2.$

If $f^n(x)=(f\circ f\circ f.....$ upto n times$)(x)$, then

$\lim\limits_{n\to\infty}\int\limits_0^1 x^{n-2}\left(f^n(x)\right)dx$ is equal to _____.

Open solution

Q350. JEE Main 2022 (Online) 25th July Morning Shift · Numerical

If $\lim\limits_{n\to\infty}\frac{(n+1)^{k-1}}{n^{k+1}}[(nk+1)+(nk+2)+\ldots+(nk+n)]$ $=33\cdot\lim\limits_{n\to\infty}\frac{1}{n^{k+1}}\cdot\left[1^k+2^k+3^k+\ldots+n^k\right]$, then the integral value of k is equal to

_____

Open solution

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Q351. JEE Main 2021 (Online) 16th March Morning Shift · **Numerical**

Let f : (0, 2) $\rightarrow$ R be defined as f(x) = $\log_2\left(1+\tan\left(\frac{\pi x}{4}\right)\right)$. Then, $\lim\limits_{n\to\infty}\frac{2}{n}\left(f\left(\frac{1}{n}\right)+f\left(\frac{2}{n}\right)+\ldots+f(1)\right)$ is equal to _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **D**	3. **B**	4. **B**	5. **B**
6. **B**	7. **B**	8. **C**	9. **C**	10. **D**
11. **C**	12. **B**	13. **C**	14. **B**	15. **B**
16. **A**	17. **D**	18. **D**	19. **B**	20. **B**
21. **C**	22. **D**	23. **B**	24. **D**	25. **A**
26. **A**	27. **A**	28. **C**	29. **C**	30. **A**
31. **B**	32. **B**	33. **B**	34. **C**	35. **B**
36. **D**	37. **C**	38. **A**	39. **B**	40. **B**
41. **B**	42. **D**	43. **A**	44. **C**	45. **B**
46. **D**	47. **A**	48. **B**	49. **B**	50. **A**
51. **A**	52. **A**	53. **C**	54. **C**	55. **C**
56. **C**	57. **A**	58. **C**	59. **C**	60. **C**
61. **B**	62. **B**	63. **B**	64. **C**	65. **D**
66. **B**	67. **B**	68. **B**	69. **C**	70. **C**
71. **B**	72. **B**	73. **B**	74. **A**	75. **C**
76. **A**	77. **B**	78. **D**	79. **A**	80. **A**
81. **A**	82. **C**	83. **A**	84. **A**	85. **A**
86. **D**	87. **B**	88. **D**	89. *Scan QR*	90. **B**
91. **A**	92. **B**	93. **D**	94. **A**	95. **A**
96. **D**	97. **B**	98. **B**	99. **C**	100. **C**
101. **D**	102. **B**	103. **A**	104. **D**	105. **C**
106. **C**	107. **D**	108. **B**	109. **C**	110. **C**
111. **B**	112. **B**	113. **A**	114. **D**	115. **D**
116. **C**	117. **C**	118. **B**	119. **C**	120. **D**
121. **B**	122. **D**	123. **A**	124. **C**	125. **B**
126. **C**	127. **B**	128. **C**	129. **B**	130. **C**
131. **B**	132. **C**	133. **A**	134. **A**	135. **A**
136. **A**	137. **D**	138. **B**	139. **B**	140. **C**
141. **D**	142. **B**	143. **A**	144. **B**	145. **A**
146. **C**	147. **C**	148. **A**	149. **D**	150. **B**

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151. **A**	152. **D**	153. **D**	154. **A**	155. **D**
156. **C**	157. **C**	158. **A**	159. **B**	160. **C**
161. **B**	162. **A**	163. **D**	164. **A**	165. **C**
166. **C**	167. **A**	168. **C**	169. **C**	170. **A**
171. **C**	172. **A**	173. **B**	174. **A**	175. **A**
176. **D**	177. **D**	178. **D**	179. **C**	180. **D**
181. **B**	182. **C**	183. **B**	184. **D**	185. **C**
186. **A**	187. **D**	188. **C**	189. **C**	190. **A**
191. **D**	192. **A**	193. **C**	194. **A**	195. **B**
196. **C**	197. **C**	198. **A**	199. **A**	200. **A**
201. **B**	202. **C**	203. **C**	204. **B**	205. **B**
206. **B**	207. **A**	208. **D**	209. **D**	210. **A**
211. **C**	212. **C**	213. **B**	214. **A**	215. **D**
216. **D**	217. **A**	218. **A**	219. **D**	220. **D**
221. **D**	222. **A**	223. **D**	224. **D**	225. **A**
226. **C**	227. **B**	228. **C**	229. **D**	230. **B**
231. **D**	232. **C**	233. **B**	234. **D**	235. **B**
236. **B**	237. **D**	238. **C**	239. **B**	240. **D**
241. **C**	242. **C**	243. **D**	244. **D**	245. **D**
246. **A**	247. **B**	248. **D**	249. **B**	250. **D**

Numerical

251. **12**	252. **192**	253. **9**	254. **210**	255. **2**
256. **64**	257. **16**	258. **36**	259. **2**	260. **9**
261. **17**	262. **8**	263. **12**	264. **64**	265. **112**
266. **32**	267. **23**	268. **65**	269. **1**	270. **48**
271. **8**	272. **219**	273. **8**	274. **15**	275. **176**
276. **27**	277. **5**	278. **155**	279. **12**	280. **6**
281. **2**	282. **41**	283. **18**	284. **575**	285. **32**
286. **6**	287. **14**	288. **13**	289. **63**	290. **1**
291. **12**	292. **20**	293. **27**	294. **22**	295. **2**
296. **104**	297. **10**	298. **385**	299. **12**	300. **24**
301. **8**	302. **5**	303. **16**	304. **3**	305. **21**
306. **6**	307. **6**	308. **1**	309. **34**	310. **5**

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311. **4**	312. **5**	313. **1**	314. **16**	315. **8**
316. **512**	317. **1**	318. **1**	319. **16**	320. **406**
321. **1**	322. **2**	323. **19**	324. **3**	325. **21**
326. **1**	327. **1.50**			

MCQ (More than One Correct Answer)

328. **C, B**				

Out of syllabus

329. **B**	330. **B**	331. **C**	332. **C**	333. **C**
334. **C**	335. **B**	336. **A**	337. **A**	338. **B**
339. **D**	340. **A**	341. **B**	342. **A**	343. **A**
344. **D**	345. **D**	346. **B**	347. **A**	348. **A**
349. **0**	350. **5**	351. **1**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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